



Security and privacy in multi-cloud and hybrid cloud environments: Challenges, strategies, and future directions

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ABSTRACT

The rapid adoption of multi-cloud and hybrid cloud environments has revolutionized modern computing by enhancing scalability, flexibility, and cost-efficiency. However, these environments introduce significant security and privacy challenges due to the distributed nature of data storage, heterogeneous infrastructures, and intercloud communications. This review comprehensively examines the critical security and privacy concerns associated with multi-cloud and hybrid cloud architectures, including data confidentiality, access control, secure communication, regulatory compliance, and emerging attack vectors such as cross-cloud threats and side-channel attacks. We analyze existing security strategies, including cryptographic techniques, identity and access management (IAM) mechanisms, AI-driven threat detection, and privacy-preserving methodologies. Furthermore, we provide a comparative evaluation of these approaches, highlighting their trade-offs in terms of security effectiveness, computational overhead, and deployment feasibility. In addition, we explore emerging trends such as post-quantum cryptography, zero-trust architectures, decentralized security frameworks, and AI-powered security automation to mitigate evolving threats. Finally, we outline open research challenges and future directions, emphasizing the need for scalable, adaptive, and regulation-compliant security solutions. This review serves as a foundation for researchers and practitioners aiming to enhance security and privacy in multi-cloud and hybrid cloud infrastructures, ensuring robust and resilient cloud computing ecosystems.

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1. Introduction

Cloud computing has fundamentally transformed the way organizations manage and deploy their IT resources, offering on-demand scalability, reduced infrastructure costs, and enhanced business agility (Aro-gundade and Palla, 2023; Segun-Falade et al., 2024; Obi et al., 2024; Liu et al., 2018; Ali et al., 2024). As enterprises increasingly pursue digital transformation, the adoption of multi-cloud and hybrid cloud strategies has accelerated, driven by the need for operational flexibility, risk mitigation, and optimal resource allocation (Gupta, 2024; Timilehin, 2024; Bieger, 2023; Sapkal et al., 2024; Kodakandla, 2024). Multi-cloud environments, using services from multiple cloud providers, offer advantages such as redundancy, vendor neutrality, and customized solutions. In contrast, hybrid cloud architectures integrate on-premises infrastructure with public and private clouds to achieve a balance between security and scalability (Zahra et al., 2024; Putri, 2025). The motivation behind adopting these strategies lies in the evolving landscape of enterprise IT demands, regulatory requirements, performance optimization goals, and disaster recovery planning (Tahmasebi, 2024; Battleson et al., 2016).

In today's era of rapid technological advancement and dynamic market conditions, organizations require agile IT frameworks capable of adapting to fluctuating workloads, diverse application requirements, and stringent security protocols (Dugbartey and Kehinde, 2025; Mishra and Agarwal, 2024). Multi-cloud and hybrid cloud environments enable companies to distribute workloads across various platforms based on performance, cost, and compliance considerations, thus ensuring operational continuity, even in the face of unexpected system failures or cyber threats (Ponnusamy and Spanner, 2023).

The evolving nature of modern applications—particularly those that incorporate layers of artificial intelligence (AI), big data analytics, and the internet of things (IoT)—necessitates that organizations adopt a diverse cloud strategy to meet the varied computing and storage demands. Using the strengths of multiple cloud providers, organizations can achieve improved performance, and redundancy against potential failures, and benefit from a competitive market that minimizes dependency on any single vendor. Furthermore, legal and regulatory requirements often require that data be stored within specific

geographic locations, making the integration of various cloud technologies advantageous for data partitioning and computation within public cloud environments (Hashem et al., 2015; Fernández et al., 2014; Yang et al., 2017). Cost considerations also play a critical role; mission-critical or sensitive applications may be hosted on private clouds to control expenses, while general-purpose workloads can be offloaded to more cost-effective public clouds. Hybrid cloud architectures facilitate seamless migration and integration between on-premises and public cloud resources, enabling enterprises to improve business agility while maintaining granular control over workload security and compliance (Mathur, 2024; Vankayalapati, 2025).

The growing use of edge computing and real-time data processing further amplifies the demand for hybrid cloud solutions, as latency-sensitive services can be executed closer to the data source, while core computations are transferred to the cloud (Wang et al., 2020a; Poojara et al., 2022; Sarkar et al., 2019). Moreover, the emergence of new technologies such as cloud computing has introduced additional security threats and challenges, necessitating the implementation of enhanced and uniform security protocols across cloud platforms. These include standardized identity and access management (IAM) methods, encryption techniques, and threat detection mechanisms (Singh et al., 2016; Praveenadevi et al., 2024). Consequently, multi-cloud and hybrid cloud architectures enable organizations to strike a balance between innovation and security, supporting advanced digital environments that promote business growth, technological progress, operational efficiency, and the protection of sensitive data in an increasingly interconnected world.

Ensuring data confidentiality, integrity, access control, and inter-cloud communication security is of paramount importance in contemporary multi-cloud environments, especially where sensitive data are fragmented and transferred across diverse platforms. The primary challenge lies in preserving the confidentiality and integrity of data during storage and transmission, as fragmented data across multiple clouds increases the risk of unauthorized access, exposure to adversarial manipulation, and potential breaches (Akinade et al., 2025; Raja et al., 2024; Jimmy, 2024; Soni and Dhurwe, 2024). This issue becomes even more critical given the massive volume of data generated in real-time cloud-based applications, where malicious alterations can compromise

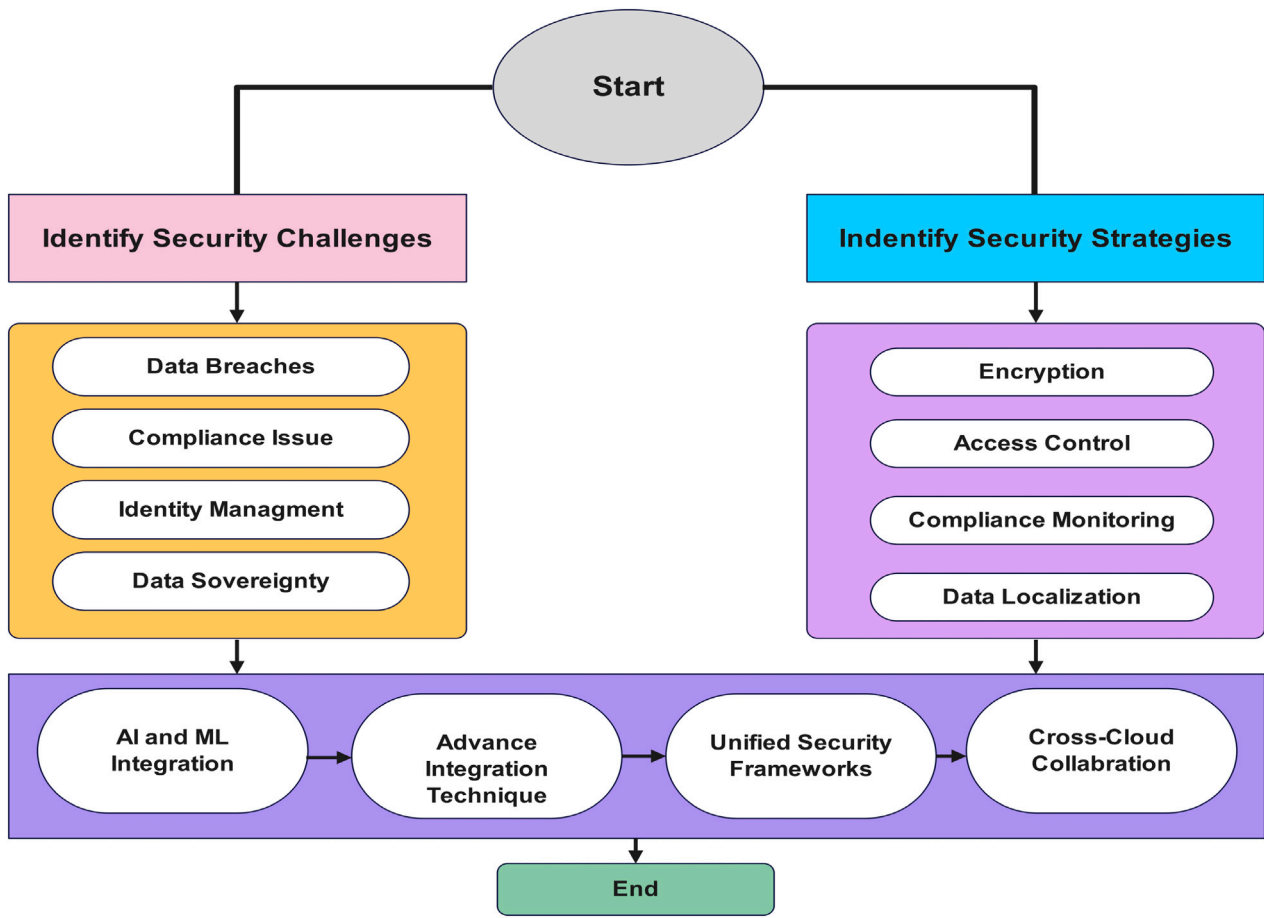


Fig. 1. Framework for addressing cloud security challenges through strategic mitigation and advanced security integration.

not just information privacy, but the entire trust infrastructure of the system (Singh et al., 2024; Ponde, 2025; Akbar et al., 2024; Shilpa, 2024). Furthermore, improper management of user identities and privileges across multiple cloud providers may result in privilege escalation, insider threats, and inconsistent enforcement of security policies (Vardha et al., 2025; Arumugam, 2024; Asha and Shanmugapriya, 2024). Finally, the adoption of hybrid and multi-cloud strategies increases the risk of intercloud vulnerabilities, including man-in-the-middle (MITM) attacks, eavesdropping, and insecure APIs, threatening data-in-transit security (AKHTAR, 2025; Reece, 2024; Routh and Ranjan, 2024).

To address these challenges, our review analyzes the layered security mechanisms that must be employed. End-to-end encryption, both at rest and in transit, using robust cryptographic schemes like AES and secure communication protocols like TLS, plays a vital role in mitigating threats related to unauthorized access and data exposure. Ensuring data integrity through hashing algorithms such as SHA-256 provides reliable verification mechanisms to detect unauthorized modifications. To strengthen access control, techniques such as single sign-On (SSO), multifactor authentication (MFA), and role-based access control (RBAC) are essential to enforce strict identity management and uphold the principle of least privilege. In the realm of inter-cloud communication, secure channels such as VPNs, encrypted APIs, and secure tunnels help safeguard data exchanges from interception and malicious intrusions. Furthermore, continuous monitoring, real-time auditing, and frequent policy updates are critical across all layers: data protection, access control, and communication to detect anomalies, revoke compromised credentials, and ensure adaptive security postures. Together, these solutions not only preserve the confidentiality and integrity of data, but also enhance the trust, resilience, and security of multi-cloud infrastructures in adversarial settings.

The (Fig. 1), is the result of our analysis presented in this review research. It was developed to illustrate the key components and interrelations identified through our comprehensive evaluation of existing security mechanisms in multi-cloud environments. We illustrate a structured approach to addressing cloud security challenges and implementing effective strategies. The figure outlines a structured approach to addressing modern security challenges in a cloud environment. It begins by identifying key security challenges such as data breaches, compliance issues, and data sovereignty, followed by corresponding strategies like encryption, access control, and data localization. The framework then highlights advanced techniques, including AI and ML integration, unified security frameworks, and cross-cloud collaboration, to enhance security measures. The process culminates in a comprehensive, integrated solution that ensures robust protection, compliance monitoring, and seamless collaboration across platforms, addressing both current and emerging threats in a scalable and efficient manner.

1.1. Scope and contributions of the review

In this subsection, we explain the scope and major contributions of the review article, as outlined below.

1.1.1. Scope of the review

This review aims to provide a comprehensive discussion of security and privacy issues in multi-cloud and hybrid cloud environments. As many organizations today operate within distributed cloud-based infrastructures, it is critical to address concerns related to security and privacy preservation. This paper identifies and categorizes various forms of security threats, vulnerabilities, and privacy risks, presenting a structured hierarchy of security challenges in cloud environments.

The study examines key aspects such as data security, identity management, inter-cloud authentication, compliance with regulatory frameworks, and the application of advanced cryptographic techniques for secure data sharing across heterogeneous cloud infrastructures. In addition, this review highlights emerging technologies and methodologies used to mitigate security risks, including zero-trust architectures, confidential computing, blockchain-based security solutions, and AI-driven threat detection mechanisms. By analyzing state-of-the-art research and industry best practices, this review seeks to bridge the gap between theoretical advancements and practical implementations, enabling organizations and cloud service providers to adopt effective security measures tailored to their specific operational needs.

1.1.2. Contributions of the review

This review provides a comprehensive synthesis of existing security and privacy frameworks, identifying gaps in current methodologies, and proposing future research directions. The primary contributions of this review are as follows:

- (1) A thorough classification of security threats and vulnerabilities specific to multi-cloud and hybrid cloud environments, outlining attack vectors and their potential impact on cloud infrastructures.
- (2) A comparative analysis of existing security solutions, evaluating their effectiveness, computational overhead, and adaptability to dynamic cloud ecosystems.
- (3) An exploration of privacy-enhancing techniques, including homomorphic encryption, secure multiparty computation, and differential privacy, to protect sensitive data while maintaining computational efficiency.
- (4) The integration of regulatory compliance considerations, mapping cloud security practices to legal frameworks such as GDPR, HIPAA, and NIST guidelines.
- (5) A discussion of emerging research challenges and open problems, offering insights into future directions for the development of resilient and scalable security architectures.

By presenting a holistic perspective on cloud security and privacy, this review serves as a foundational resource for researchers, industry practitioners, and policymakers striving to enhance the security posture of multi-cloud and hybrid cloud infrastructures.

The (Fig. 2), presents a comprehensive roadmap for a review article focused on security in multi-cloud and hybrid cloud environments. It begins with an Introduction (Section 1), outlining the scope, contributions, and research questions of the review. (Section 2) provides an overview of architectural models, performance optimization, reliability, and interoperability aspects. (Section 3) highlights key security and privacy challenges, including IAM issues, network security, threats, and compliance requirements. (Section 4) delves into the limitations of multi-cloud setups, with emphasis on interoperability, federated identity, policy normalization, and formal verification tools. (Section 5) surveys existing security and privacy strategies, followed by (Section 6), which compares these strategies across different platforms. (Section 7) Explores emerging trends and future research avenues. (Section 8) identifies open challenges, such as scalability, security-cost trade-offs, and compliance with evolving regulations. The roadmap concludes in (Section 9) with key findings and actionable recommendations to advance cloud security research and development. Each section builds logically to form a structured, in-depth analysis of the current and future cloud security landscape.

2. Overview of multi-cloud and hybrid cloud environments: Analytical approaches

In recent years, the paradigm shift from monolithic cloud deployments to heterogeneous and distributed infrastructures has catalyzed the emergence of Multi-Cloud and Hybrid Cloud environments, driven by the dual imperatives of cost-efficiency and resilience (Gupta, 2024; Timilehin, 2024). At a foundational level, we define a Multi-Cloud

Environment as a tuple $C_{multi} = \{C_1, C_2, \dots, C_n\}$, where each C_i corresponds to a distinct cloud service provider $C_i \in \mathbb{C}_{pub}$, such that $\mathbb{C}_{pub} = \{AWS, Azure, GCP, \dots\}$, with orthogonality in services, cost structures, and compliance regimes. Conversely, a Hybrid Cloud Environment is mathematically constructed as a union $C_{hybrid} = C_{priv} \cup C_{pub}$, where C_{priv} is a finite set of on-premises or private clouds with restricted access, and $C_{pub} \subseteq \mathbb{C}_{pub}$ as before. The motivation to adopt such systems comes from optimization objectives that can be formalized using utility functions. Suppose $\vec{x} \in \mathbb{R}^n$ denotes the vector of resources allocated in multiple clouds; then a generalized cost-performance trade-off function is given by $U(\vec{x}) = \alpha f_{perf}(\vec{x}) - \beta f_{cost}(\vec{x})$, where $\alpha, \beta \in \mathbb{R}^+$ are weights indicating the performance of the system and the sensitivity of the budget. In a multi-cloud scenario, we are often interested in load-balancing resource assignments subject to both capacity and cost constraints (Suresh et al., 2024; Zhang et al., 2021b; Kanbar and Faraj, 2022), forming a constrained optimization problem: $\min_{\vec{x}} f_{cost}(\vec{x})$ subject to $A\vec{x} \leq \vec{b}$, where matrix $A \in \mathbb{R}^{m \times n}$ encodes resource limits across clouds and $\vec{b} \in \mathbb{R}^m$ denotes maximum capacity per region or service type. Similarly, data migration and compliance introduce temporal dynamics (Zaki et al., 2024), modeled using differential equations $\frac{d\vec{x}}{dt} = F(\vec{x}, t)$, reflecting how service usage evolves. These dynamic systems must be evaluated under stochastic influences such as network latency, failures, or pricing volatility, necessitating probabilistic models (Cheng et al., 2018; Sarimveis et al., 2008).

Let $X \sim P(x)$ be a random variable representing cloud availability; the expected uptime of the system $\mathbb{E}[U(X)]$ can be derived from empirical distributions or modeled through a Poisson process $N(t) \sim \text{Poisson}(\lambda t)$, where λ denotes the expected number of service disruptions per unit of time. Furthermore, in hybrid deployments involving sensitive workloads, we define security exposure as an entropy function over access configurations $H(X) = -\sum_i p(x_i) \log_2 p(x_i)$, where $p(x_i)$ is the probability distribution over possible user access events. Thus, the analytical study of Multi-Cloud and Hybrid Cloud environments is inherently multidisciplinary, requiring tools from linear programming, stochastic calculus, information theory, and systems modeling to rigorously address optimization, security, and scalability (Abdo, 2021; Grandinetti, 2013). As cloud adoption becomes increasingly decentralized, understanding and formalizing these environments through precise mathematical abstractions becomes not only beneficial but essential, for designing robust and future-proof architectures (Gai, 2020).

2.1. Architectural overview and taxonomy

The architectural foundation of multi-cloud and hybrid cloud environments encapsulates a layered, distributed framework comprising diverse service models, infrastructure types, and control policies, which can be systematically modeled using graph theory, set algebra, and linear systems (Merseedi and Zeebaree, 2024; Shrivastava and Agrawal, 2024). At the core of this architecture lies the resource classification across service layers, Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS) - which can be formally represented as disjoint sets $\mathcal{R}_I, \mathcal{R}_P, \mathcal{R}_S$, respectively, such that $\mathcal{R}_{cloud} = \mathcal{R}_I \cup \mathcal{R}_P \cup \mathcal{R}_S$, with $\mathcal{R}_I \cap \mathcal{R}_P = \emptyset$, ensuring nonoverlapping resource semantics. Each service model is hierarchically dependent, that is, there exists a function mapping between the layers $\phi: \mathcal{R}_I \rightarrow \mathcal{R}_P \rightarrow \mathcal{R}_S$, modeled as a directed acyclic graph (DAG) $G = (V, E)$, where each node $v \in V$ denotes a service or component, and the directed edge $(v_i, v_j) \in E$ represents a dependency path.

From a topological standpoint, multi-cloud and hybrid cloud infrastructures can be visualized as a multi-graph of interconnected cloud nodes, where the graph $\mathcal{G}_{cloud} = (V, E)$ contains vertices $V = \{v_1, v_2, \dots, v_n\}$ that represent virtualized data centers or resource clusters from different providers (public or private), and edges E signify bandwidth-constrained communication links. We assign weights w_{ij} to each edge $e_{ij} \in E$ to represent the cost or latency of communication,

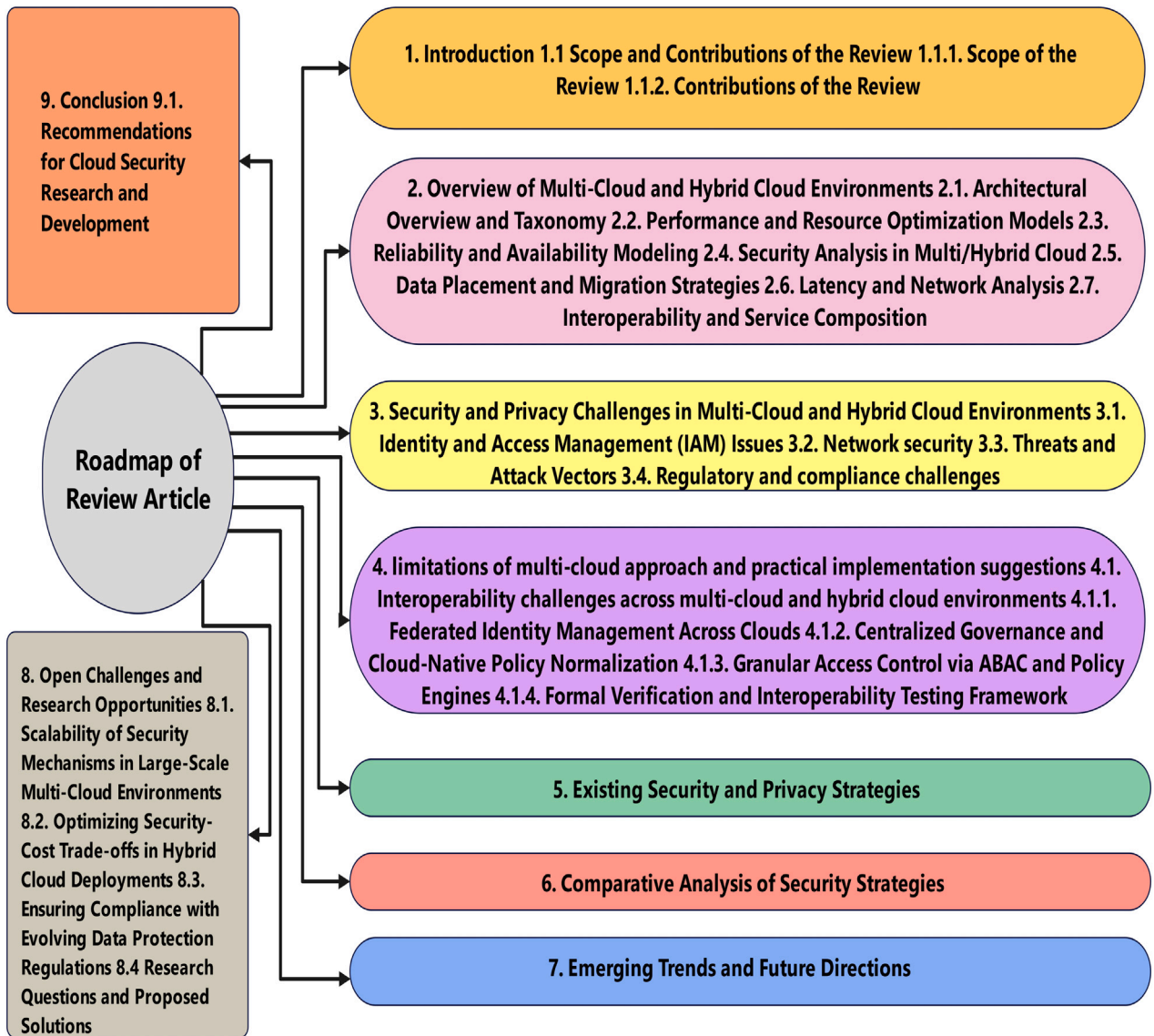


Fig. 2. Roadmap of the review article.

thus forming a weighted graph G_{uc} . The connectivity matrix $A = [a_{ij}]$ of the graph is binary if $a_{ij} = 1$ indicates a direct communication path between v_i and v_j , and zero otherwise; however, in performance modeling, we generalize this into a weighted adjacency matrix $W = [w_{ij}]$ where $w_{ij} \geq 0$ denotes latency, cost or reliability metrics. The degree centrality of a node v_k is defined as $D(v_k) = \sum_{j=1}^n a_{kj}$, reflecting the number of cloud systems with which it directly interacts, which is critical for understanding orchestration overhead in complex architectures.

The taxonomy of multi-cloud environments can be divided into four orthogonal dimensions: provider heterogeneity, service abstraction, orchestration mechanisms, and compliance boundaries (Tomarchio et al., 2020; Brabra, 2020; Weerasiri et al., 2017). Let us define a configuration space $\Omega = \mathcal{P} \times \mathcal{S} \times \mathcal{O} \times \mathcal{C}$, where $\mathcal{P}$ is the set of cloud providers $\{P_1, P_2, \dots, P_m\}$, $\mathcal{S}$ is the set of abstracted services, $\mathcal{O}$ is the orchestration policy domain and $\mathcal{C}$ is the compliance rule set. Each instance of the system $\omega \in \Omega$ can be evaluated as a tensor product $\omega = P_i \otimes S_j \otimes O_k \otimes C_l$, capturing the multidimensional classification of cloud deployment strategies. This tensorial formulation is pivotal in systems synthesis and enables algebraic manipulation for constraint satisfaction, security modeling, and deployment optimization.

In hybrid cloud architecture, the communication and integration layer is paramount and is governed by federated identity management,

workload portability, and secure tunneling mechanisms (Anh, 2024). Mathematically, this can be modeled using interoperability matrices $I = [i_{ab}]$, where $i_{ab} = 1$ indicates that the service a of the private cloud C_{priv} is interoperable with the service b of the public cloud C_{pub} . Interoperability is often constrained by API semantics and vendor-specific bindings, which can be described by defining compatibility functions $\kappa(a, b) = 1$ if $API_a \approx API_b$, and 0 otherwise. This gives rise to an interoperability graph $G_{interop}$, and finding the largest clique in $G_{interop}$ gives the maximum interoperable subsystem, an NP-complete problem that can be heuristically solved using graph coloring algorithms.

Further, the cloud control and orchestration plane can be abstracted as a finite-state machine (FSM) $\mathcal{M} = (\mathcal{S}, \Sigma, \delta, s_0, F)$, where $\mathcal{S}$ is the set of orchestration states (e.g., "provisioned", "scaling", "decommissioning"), Σ is the input set (e.g., demand surge, policy violation), δ is the transition function, and s_0 and F are the initial and terminal states respectively. This FSM helps formalize dynamic behavior such as auto-scaling, fault tolerance triggers, and load redistribution in response to external stimuli.

Finally, in the taxonomy of workloads and data types, we define a workload as a function $W : T \rightarrow \mathbb{R}^+$ mapping time to resource demand, and classifying workloads into deterministic (periodic or real-time) and stochastic (bursty or ad-hoc) categories. The workload vector

$\vec{w} = [w_1(t), w_2(t), \dots, w_n(t)]$ on cloud nodes n forms the input to a system-wide scheduler, where the optimization objective is minimizing the total cost $\sum_{i=1}^n c_i(w_i)$ under SLA and latency constraints $\mathbb{E}[L_i] \leq L_{\max}$. These objectives are typically modeled using queueing networks, with each cloud node modeled as a server $M/M/1$ or $M/G/1$ queue. The taxonomy is further enriched with notions of data sovereignty and regulatory domains, modeled as partitioned subsets $D_i \subseteq \mathbb{D}$ (the universal data space), constrained by $\forall D_i, \exists R_j : D_i \rightarrow \text{only regions in } R_j$, enforcing geofencing and jurisdictional compliance via propagation of logical constraints.

2.2. Performance and resource optimization models

The performance and resource optimization in Multi-Cloud and Hybrid Cloud environments is fundamentally governed by a set of mathematical models involving linear programming, queueing theory, stochastic optimization, and control systems (Ciavotta et al., 2017; Khaleel, 2024; Ma et al., 2024; Nawrocki and Smendowski, 2025). These environments can be viewed as distributed systems that seek to minimize operational costs while maximizing throughput, resilience, and compliance under dynamic constraints (Kommera, 2013; Amiri et al., 2023). We define the state of the global system as a tuple $S(t) = \langle \vec{r}(t), \vec{w}(t), \vec{u}(t), \vec{l}(t) \rangle$, where $\vec{r}(t)$ is the vector of real-time resource availability, $\vec{w}(t)$ the workload demands, $\vec{u}(t)$ the current utilization per node and $\vec{l}(t)$ the latency per task. At any time t , the system aims to solve an optimization problem of the form:

$$\min_{\vec{x}(t)} \left[\sum_{i=1}^n C_i(x_i(t)) + \lambda_1 \sum_{i=1}^n L_i(x_i(t)) - \lambda_2 \sum_{i=1}^n T_i(x_i(t)) \right] \quad (1)$$

Subject to $A\vec{x}(t) \leq \vec{b}(t)$, $\vec{x}(t) \in \mathbb{R}_+^n$, where $C_i(\cdot)$ is the cost function for node i , $L_i(\cdot)$ is the latency, $T_i(\cdot)$ is the throughput, and $\lambda_1, \lambda_2 \in \mathbb{R}^+$ are weight coefficients that balance the delay and performance objectives. These formulations are embedded within multi-objective linear programming (MOLP) models or constrained convex optimization problems, depending on the curvature of cost and delay functions (Karimi et al., 2022; Shir and Emmerich, 2024; Klamroth and Jørgen, 2007).

In the workload modeling layer, let $W_i(t) \sim \mathcal{P}_i$ represent the workload at node i governed by a stochastic process $\mathcal{P}_i$. The temporal behavior of workloads across clouds is best modeled as a stochastic process $\{W(t)\}_{t \geq 0}$, often assumed to be Markovian or piecewise stationary. For predictable workloads, a deterministic function $w_i(t) = a_i \sin(\omega_i t + \phi_i) + b_i$ can be used to model cyclic usage. For bursty workloads, Poisson or self-similar processes $N_i(t) \sim \text{Poisson}(\lambda_i)$ or H -fractal models with Hurst parameter $H \in (0.5, 1)$ are more appropriate. The objective is to calculate the optimal allocation vector $\vec{x}^* \in \mathbb{R}^n$ that maps workloads to resources such that service-level agreements (SLAs) are met and the energy consumption is minimized.

In resource provisioning and elasticity modeling, we represent the available capacity of each node i as $r_i(t)$, with a provisioning function $P_i(t)$ mapping the projected load to the provisioned resources. This can be formulated through a feedback control loop:

$$r_i(t+1) = r_i(t) + \Delta r_i(t), \quad \text{where} \quad \Delta r_i(t) = K_p e_i(t) + K_d \frac{d}{dt} e_i(t) \quad (2)$$

Here, $e_i(t) = w_i(t) - u_i(t)$ is the error signal, and K_p, K_d are the proportional and derivative gains in a PID controller used for auto-scaling policies. These dynamical systems can be analyzed using Laplace transforms to evaluate stability and overshoot under variable workloads (Marx et al., 2011). This provides a control-theoretic formulation of elasticity that reacts to exogenous input perturbations (i.e., sudden spikes in demand).

For task scheduling and load balancing, we define a mapping function $\sigma : T \rightarrow C$ where task T_i is mapped to cloud C_j , subject to minimization of expected latency. Let ℓ_{ij} denote the latency for task i on node j , then the scheduling problem becomes:

$$\min_{\sigma} \sum_{i=1}^m \ell_{i\sigma(i)} \quad \text{subject to} \quad \sum_{i: \sigma(i)=j} w_i(t) \leq r_j(t), \quad \forall j \quad (3)$$

This is a classic minimum-cost bipartite matching problem with additional capacity constraints and can be solved using Hungarian algorithms, greedy heuristics, or LP relaxation (Horváth and Kis, 2016; Abeywickrama et al., 2021). In multi-tenant systems, tasks are prioritized using priority queues Q_k and modeled using queueing networks, where each cloud provider is represented as a queue $M/M/c$ or $G/G/1$, and system utilization is given by $\rho_i = \frac{\lambda_i}{\mu_i c_i}$, where λ_i is the task arrival rate, μ_i is the service rate, and c_i is the number of servers.

The energy and cost optimization objective is formalized using a weighted linear combination:

$$J = \sum_{i=1}^n (\alpha_i E_i + \beta_i C_i) \quad (4)$$

where $E_i = \int_0^T P_i(t) dt$ is the energy consumption (with $P_i(t)$ being the power drawn at time t), and C_i is the monetary cost function based on the computing, memory, bandwidth, and storage usage. Using Lagrangian multipliers λ , constrained optimization under SLA thresholds $S_i \leq S_{\max}$ is achieved by solving:

$$\mathcal{L}(\vec{x}, \lambda) = J(\vec{x}) + \lambda (S_i(\vec{x}) - S_{\max}) \quad (5)$$

The Karush–Kuhn–Tucker (KKT) conditions are checked for optimality (Clemptner, 2016). Multi-cloud environments introduce pricing heterogeneity $\pi_i(t)$, often modeled as a non-stationary time series with abrupt shifts, prompting the use of online convex optimization (OCO) where decisions x_t are made sequentially to minimize regret:

$$\text{Regret}_T = \sum_{i=1}^T f_i(x_i) - \min_{x \in \mathcal{X}} \sum_{i=1}^T f_i(x) \quad (6)$$

With $f_i(x)$ being the cost at time t , and $\mathcal{X}$ the feasible set. Techniques like Mirror Descent (Orabona et al., 2015), Follow-The-Regularized Leader (FTRL) (Chen and Orabona, 2023), and Gradient Descent with Projection are used to adapt resource allocation underpricing volatility (Chen et al., 2017).

In compliance-aware and location-sensitive optimization, data placement is modeled as a bin-packing problem under constraints. Define $D = \{d_1, d_2, \dots, d_k\}$ as data items with location tags $L(d_i) \subset \mathcal{R}$, and clouds as bins B_j with storage capacity S_j . Each assignment must satisfy:

$$\sum_{d_i \rightarrow B_j} \text{size}(d_i) \leq S_j \quad \text{and} \quad L(d_i) \in R_j \quad (7)$$

This becomes a multi-dimensional bin-packing problem with side constraints, which is NP-hard and approached via approximation algorithms, heuristics, or integer programming (Ali et al., 2025b). The placement cost $C_{\text{place}}(d_i, B_j)$ is often modeled as a linear function of data transfer time, jurisdictional penalty, and latency, adding another dimension to optimization.

Lastly, failure resilience and reliability modeling use Markov chains and stochastic reliability theory (El-Awady and Ponnambalam, 2021). Let each cloud node be a two-state Markov model with transition rates λ (failure) and μ (repair). The steady-state availability $A = \frac{\mu}{\lambda + \mu}$ and the mean time to failure (MTTF) are given by $\text{MTTF} = \frac{1}{\lambda}$. For a system of n replicated services, the general availability of the system under k -out-of- n redundancy is:

$$A_{\text{sys}} = \sum_{i=k}^n \binom{n}{i} A^i (1-A)^{n-i} \quad (8)$$

Which forms the reliability envelope for the delivery of the service. Incorporating this into the optimization framework ensures that performance targets are met under probabilistic fault models.

2.3. Reliability and availability modeling

The modeling of reliability and availability in Multi-Cloud and Hybrid Cloud environments requires a mathematically grounded approach due to the stochastic nature of failures, diverse geographical

deployments, and heterogeneous resource architectures (Alam et al., 2021). In such distributed environments, a cloud service is generally modeled as a series of interdependent subsystems across multiple cloud providers (Toosi et al., 2014; Ghahramani et al., 2017). Let us define the system as a set $\mathcal{N} = \{n_1, n_2, \dots, n_k\}$, where each node n_i denotes a cloud component or service module. The fundamental metrics, availability A and reliability R , are classically defined as probabilistic functions of system uptime. The instantaneous availability of node n_i is given by $A_i(t) = \Pr[n_i \text{ is operational at time } t]$, while the reliability is defined over a time interval $[0, t]$ as $R_i(t) = \Pr[n_i \text{ does not fail in } [0, t]]$. A common assumption is that failures and repairs are memoryless processes, rendering each node a two-state continuous-time Markov chain (CTMC), with transition rates λ_i (failure rate) and μ_i (repair rate). The steady-state availability A_i is derived from the limiting probabilities as $A_i = \frac{\mu_i}{\lambda_i + \mu_i}$, and the mean time to failure (MTTF) and the mean time to repair (MTTR) are given by $\text{MTTF}_i = \frac{1}{\lambda_i}$, $\text{MTTR}_i = \frac{1}{\mu_i}$, respectively. For a serially connected system where all components must function for the service to be available, the availability of the system is the product: $A_{\text{sys}}^{(\text{series})} = \prod_{i=1}^k A_i$. In contrast, for parallel systems with redundancy where at least one component must function, the system availability is: $A_{\text{sys}}^{(\text{parallel})} = 1 - \prod_{i=1}^k (1 - A_i)$.

To model multi-cloud fault-tolerant deployments (Neto et al., 2019), consider a configuration in which a service is replicated across n clouds and can tolerate up to $n-k$ failures. This forms a k -out-of- n redundancy model, with availability calculated as:

$$A_{k/n} = \sum_{i=k}^n \binom{n}{i} A^i (1-A)^{n-i} \quad (9)$$

where A is the availability of a single cloud node. This binomial distribution captures the probability that at least k of n replicas are operational. A higher n improves fault tolerance but increases costs and complexity. For systems incorporating redundancy, we assign each node n_i a location ℓ_i and compute location-dependent failure correlation using a joint distribution $f(\lambda_i, \lambda_j | \text{distance}(\ell_i, \ell_j))$, since disasters or network outages can cause correlated failures in nearby data centers. The impact of such correlated risks is modeled using copula functions, or more simply, a correlation matrix $\Sigma \in \mathbb{R}^{n \times n}$, where each entry $\Sigma_{ij} = \rho_{ij}$ represents the Pearson correlation coefficient of failures between n_i and n_j .

The availability-aware deployment of applications then becomes a constrained optimization problem. Let $x_i \in \{0, 1\}$ indicate whether a component is deployed in the cloud i , and let A_i denote its availability. The objective is to maximize the overall availability of the system while satisfying the cost and latency constraints:

$$\max_{\vec{x}} A_{\text{sys}}(\vec{x}) = \sum_{i \in \mathcal{V}} A_i(\vec{v}) \cdot \prod_{i=1}^n x_i^{v_i} (1-x_i)^{1-v_i} \quad (10)$$

Subject to: $\sum_{i=1}^n c_i x_i \leq C_{\text{max}}$ (cost constraint), $\sum_{i=1}^n \ell_i x_i \leq \ell_{\text{max}}$ (latency constraint), where $\mathcal{V} \subseteq \{0, 1\}^n$ represents valid deployment vectors satisfying system-level logic (e.g., quorum policies), c_i is the cost of using cloud i , and ℓ_i is the average latency. Because the objective function is a nonlinear function of binary variables, this becomes a binary nonlinear programming (BNLP) problem (Murray and Ng, 2010), often approached via genetic algorithms (Kramer and Kramer, 2017), branch-and-bound (Morrison et al., 2016), or Lagrangian relaxation techniques (Anderson et al., 2025; Zhu et al., 2025).

For virtual machine (VM) availability modeling (Silva and Callou, 2025), consider $V = \{v_1, v_2, \dots, v_m\}$ as VMs hosted on n clouds. The failure of a VM v_j depends on its hosting cloud and the underlying hypervisor. Define:

$$A_{v_j} = A_{\text{cloud}(i)} \cdot A_{\text{hypervisor}(i)} \cdot A_{\text{hardware}(i)} \quad (11)$$

For $v_j \in V$ hosted on cloud node i . In practice, hypervisor reliability $A_{\text{hypervisor}}$ can degrade due to resource contention, modeled by queueing systems like $M/M/1$ or $M/G/1$ queues. Let $\rho = \lambda/\mu$ be the

traffic intensity, then availability with queueing delay tolerance can be approximated as:

$$A_{\text{queue}} = 1 - \Pr(\text{response time} > \tau) \approx 1 - e^{-\mu(1-\rho)\tau} \quad (12)$$

where τ is the SLA latency threshold. This shows that availability is inversely related to system congestion, linking availability modeling to queueing theory.

Finally, to ensure dynamic availability under failover, consider a Markov Decision Process (MDP) (Udkar et al., 2025), where states s_t represent system health, actions a_t correspond to migration or failover operations, and rewards $R(s_t, a_t)$ capture recovery time and cost penalties. The optimal availability strategy is derived from the Bellman equation:

$$V(s_t) = \max_{a_t} \left[R(s_t, a_t) + \gamma \sum_{s_{t+1}} P(s_{t+1} | s_t, a_t) V(s_{t+1}) \right] \quad (13)$$

where $\gamma \in [0, 1]$ is a discount factor. Solving this yields a policy π^* for proactive failover and redundancy, improving the expected availability over time.

2.4. Security analysis in multi/hybrid cloud

Security in Multi-Cloud and Hybrid Cloud environments must be rigorously analyzed due to their inherent heterogeneity, decentralized control, and dynamic resource provisioning (Chauke et al., 2025). These environments introduce a larger threat surface due to inter-provider interactions, varied trust assumptions, and shared virtual infrastructures (Anh, 2024). To model the security of such environments, we define the global cloud security state as $S_{\text{sec}}(t) = \langle \mathcal{A}(t), \mathcal{P}(t), \mathcal{K}(t), \mathcal{T}(t) \rangle$, where $\mathcal{A}(t)$ is the access state, $\mathcal{P}(t)$ the policy state, $\mathcal{K}(t)$ the key management status, and $\mathcal{T}(t)$ the active threat set at time t . Each of these components can be individually modeled using mathematical constructs to represent evolving risk, access violations, and data compromise. To begin with, access control models such as Role-Based Access Control (RBAC) are formalized using binary matrices (Beniwal et al., 2025). Let $A = [a_{ij}] \in \{0, 1\}^{m \times n}$ be the access matrix where $a_{ij} = 1$ if user u_i has access to resource r_j , and zero otherwise. Let $R \in \mathbb{R}^{n \times k}$ map each resource to a sensitivity level or clearance class. Then, the access violation condition is modeled by $V = A \cdot R$, where $V_{ik} > \theta_k$ denotes a policy violation for the threshold $\theta_k \in \mathbb{R}^+$. For multi-cloud identity federation, let $\mathcal{U} = \{u_1, u_2, \dots, u_m\}$ be a universal user set and $\mathcal{D}_i \subset \mathcal{U}$ the identity domains managed by the provider i . The intersection $\bigcap_{i=1}^n \mathcal{D}_i \neq \emptyset$ requires federated trust between providers. We define trust functions $T : \mathcal{D}_i \times \mathcal{D}_j \rightarrow [0, 1]$ that represent the confidence level of domain i that delegates authentication to domain j . A trust graph $G_T = (V_T, E_T, w)$, where $V_T = \{\mathcal{D}_1, \dots, \mathcal{D}_n\}$ and $w(i, j) = T(i, j)$, provides a weighted structure to analyze federated identity trust propagation using path composition or shortest path-based attenuation.

In terms of cryptographic enforcement, consider the encryption state space $\mathcal{E} = \{E_1, \dots, E_m\}$, where each E_i represents an encryption scheme (e.g., AES, RSA, ECC, lattice-based), and each file f_j stored in the cloud is associated with a cryptographic tuple $\kappa_j = (E_j, K_j, D_j)$, where K_j is the key and $D_j \subseteq \mathcal{U}$ the authorized decryption domain. Secure data retrieval requires $\forall u_i \in D_j, \text{PrivKey}(u_i) \in \text{Dom}(K_j)$. In multi-cloud contexts, the key management function becomes a distribution problem, often modeled using secret sharing schemes such as Shamir's (t, n) threshold model: a secret $S \in \mathbb{F}_q$ is divided into n shares so that any t subset can reconstruct S , but any subset of size $< t$ does not provide information. Let shares $S_1, \dots, S_n$ be distributed over n clouds. The adversarial advantage is expressed by mutual information $I(S; \{S_i\}_{i \in C_a}) = 0$ if $|C_a| < t$, preserving perfect secrecy under coalition attacks. This model ensures zero knowledge leakage in the face of partial cloud compromise, particularly relevant in hybrid clouds with partially trusted private zones.

To model the dynamics of the attack surfaces and the evolution of the threat, we introduce a stochastic threat process $\{T(t)\}_{t \geq 0}$, where

$T(t) \in \mathbb{N}^+$ represents the number of active threats at time t . These threats are assumed to arrive according to a nonhomogeneous Poisson process (NHPP) with a time-dependent arrival rate $\lambda(t)$, such that the probability of observing k new threats in a small interval Δt is governed by the Poisson distribution: $\mathbb{P}(T(t + \Delta t) - T(t) = k) = \frac{(\lambda(t)\Delta t)^k}{k!} e^{-\lambda(t)\Delta t}$. The resilience of the system is quantified through a security control function $\sigma : T(t) \mapsto \mathbb{R}^+$, which assigns a cost to the defense resources required for mitigating the current threat level. This cost is captured by the control cost function $C(t) = \sigma(T(t))$, and the main objective is to optimize $C(t)$ over time to maintain a balance between security effectiveness and resource expenditure:

$$\min_{\sigma} \mathbb{E} \left[\int_0^T C(\sigma(T(t))) dt + \alpha \cdot L(T(T)) \right] \quad (14)$$

where $L(\cdot)$ represents breach-induced loss and α is a risk-aversion coefficient. This formulation leads to applications of stochastic control theory in cloud defense. Furthermore, game-theoretic models can be applied by constructing a zero-sum matrix game $G = (A, D)$, where attackers choose attack vectors $a_i \in A$, and defenders select defense strategies $d_j \in D$. The payoff matrix $M = [m_{ij}]$ encodes the expected impact or loss: $m_{ij} = \mathbb{E}[\text{damage}(a_i, d_j)]$. The saddle-point strategy (a^*, d^*) is obtained when:

$$\max_{a \in A} \min_{d \in D} m_{ad} = \min_{d \in D} \max_{a \in A} m_{ad} \quad (15)$$

Which leads to equilibrium policies under adversarial settings in multi-cloud deployments.

From an information-theoretic perspective (Tan and Huang, 2025; Kazempour et al., 2025), data exposure is quantified by the Shannon entropy $H(X) = -\sum_i p(x_i) \log_2 p(x_i)$, where X denotes access or leak events. Higher entropy implies lower predictability, and thus improved obfuscation. In contrast, mutual information $I(X; Y) = H(X) - H(X|Y)$ captures information leakage through correlated cloud behaviors or side-channel attacks. Secure configurations must enforce $I(X; Y) \approx 0$ for any $Y \in$ observable parameters. Hybrid clouds are particularly susceptible to inference attacks (Alejandra and Elena, 2025), where correlations between public cloud access logs and private workload patterns can lead to deanonymization. Thus, differential privacy mechanisms must be implemented, enforcing:

$$\frac{\Pr[\mathcal{M}(D_1) = o]}{\Pr[\mathcal{M}(D_2) = o]} \leq e^\epsilon \quad (16)$$

For all outputs o and adjacent datasets D_1, D_2 , with ϵ denoting the privacy budget. This ensures robustness against statistical inference even when partial observations are available across clouds.

Finally, multi-cloud compliance with regulatory frameworks such as GDPR, HIPAA, or PCI-DSS is modeled using compliance matrices $C = [c_{ij}]$, where $c_{ij} = 1$ if the resource r_j in the region i satisfies the compliance constraint κ_j . The legal feasibility of a data flow graph G_{flow} is checked by ensuring that every edge $(r_i, r_j) \in E_{flow}$ satisfies $c_{i\kappa_j} = 1$, that is, the regulatory propagation is preserved. Violations are assigned compliance risk vectors and integrated into risk-aware optimization through penalty functions. Thus, security modeling in multi-cloud and hybrid cloud environments is inherently multilayered, requiring stochastic, algebraic, and game-theoretic constructs to holistically address confidentiality, integrity, availability, and compliance under adversarial and uncertain operational conditions.

2.5. Data placement and migration strategies

In multi-cloud and hybrid cloud environments, data placement and migration are critical factors that significantly impact performance (Sadique et al., 2025), cost-efficiency, latency, and regulatory compliance. The core objective is to identify optimal storage locations for data chunks across multiple cloud providers and to design effective migration strategies that adapt to dynamic workloads and policy constraints. The infrastructure is modeled as a directed graph

$G = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N} = \{n_1, n_2, \dots, n_k\}$ represents the set of cloud nodes (data centers), and $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ denotes the network interconnections, each characterized by bandwidth B_{ij} and latency L_{ij} . Given a dataset $D = \{d_1, d_2, \dots, d_m\}$, a placement function $\phi : D \rightarrow \mathcal{N}$ is defined, where $\phi(d_i) = n_j$ indicates that data chunk d_i is stored at node n_j . The total placement cost function can then be formulated based on this mapping, incorporating factors such as storage cost, transfer cost, and access latency:

$$C_{\text{place}} = \sum_{i=1}^m \left(\alpha_i \cdot \text{storage}_{\phi(d_i)} + \beta_i \cdot \text{access}_{\phi(d_i)} + \gamma_i \cdot \text{compliance}_{\phi(d_i)} \right) \quad (17)$$

where $\alpha_i, \beta_i, \gamma_i \in \mathbb{R}^+$ are weight parameters capturing the cost per unit for storage, access latency, and legal compliance, respectively. The compliance term introduces a constraint set C_i for each data d_i , such that $\phi(d_i) \in C_i$ must be satisfied to meet legal requirements (e.g., data sovereignty). Access latency is modeled using the expected round trip time $\mathbb{E}[L_{ij}]$, derived from probabilistic demand distributions. Let $\lambda_i(n_j)$ denote the access rate of user i to data located at n_j , then the expected latency per data object is:

$$\mathbb{E}[T_{d_i}] = \sum_{j=1}^{|\mathcal{N}|} \lambda_i(n_j) \cdot L_{u_i, n_j} \quad (18)$$

where u_i is the location of the user and L_{u_i, n_j} the network delay from the user u_i to the node n_j . Aggregating across all users yields the total latency footprint of the placement:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^m \mathbb{E}[T_{d_i}] \quad (19)$$

To minimize both cost and latency under capacity constraints, define the multi-objective optimization problem:

$$\begin{aligned} \min_{\phi} \quad & w_1 C_{\text{place}} + w_2 \mathcal{L}_{\text{total}} \\ \text{subject to} \quad & \sum_{d_i: \phi(d_i)=n_j} s_i \leq S_j, \quad \forall n_j \in \mathcal{N} \\ & \phi(d_i) \in C_i, \quad \forall d_i \in D \end{aligned} \quad (20)$$

where s_i is the size of the data d_i , S_j is the storage capacity of node n_j , and w_1, w_2 are balancing weights. For data migration, consider a time-indexed migration schedule $M(t) = \{(d_i, n_{src}, n_{dst}, t_i)\}$, denoting data d_i moving from n_{src} to n_{dst} at time t_i . The migration cost function is:

$$C_{\text{mig}} = \sum_{(d_i, n_{src}, n_{dst}, t_i) \in M(t)} \left(\eta_i \cdot s_i \cdot \frac{1}{B_{src, dst}} + \delta_i \cdot L_{src, dst} \right)$$

where η_i is the cost per unit of bandwidth and δ_i the cost associated with the delay or interruption of the service. The bandwidth term is inverted to reflect the time-dependent transmission cost. Migration operations must satisfy non-interference constraints: define $\mathcal{M}(t)$ as a migration set active at time t ; then:

$$\sum_{(d_i, n_j, n_k, t_i) \in \mathcal{M}(t)} s_i \leq \beta_{jk} \cdot B_{jk} \cdot \Delta t, \quad \forall (n_j, n_k) \in \mathcal{E} \quad (21)$$

Which ensures total data traffic on a link does not exceed the allowable bandwidth over the interval Δt . Furthermore, the availability constraint is enforced using binary decision variables $x_{ij}(t) \in \{0, 1\}$, where $x_{ij}(t) = 1$ implies that d_i is available at node n_j at time t . For replication-aware systems, one requires:

$$\sum_{j=1}^{|\mathcal{N}|} x_{ij}(t) \geq \rho_i, \quad \forall i, t \quad (22)$$

With $\rho_i \in \mathbb{N}$ denoting the replication factor for data d_i , ensuring fault tolerance. These binary variables can be solved using integer linear programming (ILP) or relaxed to continuous optimization using Lagrangian multipliers to yield a tractable convex formulation. Furthermore, in the hybrid cloud context, migration decisions are subject to

cost differences between private and public clouds. Let μ_{priv}, μ_{pub} be the unit storage costs and σ_i be the security sensitivity score of the data d_i . Then the cloud assignment penalty is:

$$\Pi_i = \sigma_i \cdot \mathbf{1}_{\phi(d_i) \in \text{Public}} + (\mu_{priv} - \mu_{pub}) \cdot s_i \quad (23)$$

Which discourages high-sensitivity data from being placed in less secure public environments unless offset by substantial economic gain. For data-intensive applications, the data affinity matrix $A = [a_{ij}] \in \mathbb{R}^{m \times m}$, where a_{ij} represents the co-access frequency of d_i and d_j , is used to co-locate frequently accessed data. The optimization objective then includes the term:

$$C_{aff} = \sum_{i,j=1}^m a_{ij} \cdot d_{ij}, \quad d_{ij} = \mathbb{I}_{\phi(d_i) \neq \phi(d_j)} \quad (24)$$

where $\mathbb{I}$ is the indicator function that penalizes the separation of data frequently accessed by co-access. By minimizing C_{aff} , the systems achieve data locality and reduce the consumption of bandwidth between clouds. Lastly, migration decisions must account for temporal demand patterns. Suppose $\lambda_i(t)$ is the request intensity for d_i at time t ; then the migration triggering function can be defined as:

$$\psi(d_i, t) = \begin{cases} 1 & \text{if } \lambda_i(t) > \tau_i \text{ and } \phi(d_i) \notin \arg \min_{n_j} L_{u_i, n_j} \\ 0 & \text{otherwise} \end{cases} \quad (25)$$

where τ_i is a threshold, and u_i the dominant access location. This enables adaptive migration, ensuring that the data follows the user workload distribution over time. Through the integration of these mathematical models, multi-cloud and hybrid systems can achieve optimized data placement and efficient, policy-compliant, and cost-effective migration, thus maximizing operational efficiency under dynamic workloads and regulatory constraints.

2.6. Latency and network analysis

In multi-cloud and hybrid cloud environments, latency and network analysis play an essential role in ensuring optimal Quality of Service (QoS), performance predictability, and efficient workload distribution (Mia and Chloe, 2024; Waseem et al., 2024). Latency, generally defined as the time delay between a client's request and the server's response (Sevencaan, 2024), is influenced by both network-level metrics such as propagation, transmission, queueing delays, and infrastructure-level choices such as data distribution between clouds and service location. Let us model the system as a directed multigraph $G = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N} = \{n_1, \dots, n_k\}$ represents distributed cloud nodes (data centers) and $\mathcal{E}$ denotes the inter-node communication links. For each edge $e_{ij} \in \mathcal{E}$, we define its latency L_{ij} as composed of multiple components:

$$L_{ij} = L_{ij}^{\text{prop}} + L_{ij}^{\text{trans}} + L_{ij}^{\text{queue}} + L_{ij}^{\text{proc}} \quad (26)$$

where $L_{ij}^{\text{prop}} = \frac{d_{ij}}{v}$ (with d_{ij} as physical distance and v as signal propagation speed), $L_{ij}^{\text{trans}} = \frac{S}{B_{ij}}$ (with S being the size of the packet and B_{ij} the bandwidth), L_{ij}^{queue} being the dynamic queueing delay depending on network congestion, and L_{ij}^{proc} representing computational delays at the routing nodes. The total expected latency between a user located at the point u and a service hosted at node n_j is therefore:

$$\mathbb{E}[T_u] = \sum_{j=1}^{|\mathcal{N}|} P_u(n_j) \cdot (L_{u, n_j} + L_{n_j}^{\text{server}}) \quad (27)$$

where $P_u(n_j)$ denotes the access probability of the user u to node n_j , and $L_{n_j}^{\text{server}}$ is the response time at node n_j . In stochastic network settings, where latency may vary with traffic and time, we can use a Markov chain model for link state evolution. Let the network link e_{ij} exist in

one of several states $\{s_1, s_2, \dots, s_r\}$ each with latency $\ell_{ij}^{(s)}$, and define the transition matrix $\mathbf{P}_{ij} \in \mathbb{R}^{r \times r}$ such that:

$$\mathbb{E}[L_{ij}] = \sum_{s=1}^r \pi_s \cdot \ell_{ij}^{(s)} \quad (28)$$

where π_s is the steady-state probability vector of the Markov chain. Furthermore, to model end-to-end latency, we consider a path $p_{u \rightarrow j} \in \mathcal{P}$, a set of consecutive links from user u to cloud node n_j , and define the path latency:

$$L_{p_{u \rightarrow j}} = \sum_{(i,k) \in p_{u \rightarrow j}} L_{ik} \quad (29)$$

Then, the minimal expected latency over all paths is given by:

$$L_u^{\min} = \min_{p_{u \rightarrow j} \in \mathcal{P}} L_{p_{u \rightarrow j}} \quad (30)$$

This is efficiently computed using Dijkstra's algorithm extended with dynamic latency weights. In hybrid clouds, cross-cloud traffic introduces additional latency due to heterogeneity in peering agreements and bandwidth asymmetry (Zahra et al., 2024). Define $\Omega_{i,j}(t)$ as the dynamic latency between clouds i and j at time t ; then the expected inter-cloud latency variance is:

$$\sigma_{\text{inter}}^2 = \frac{1}{T} \int_0^T (\Omega_{i,j}(t) - \bar{\Omega}_{i,j})^2 dt \quad (31)$$

where $\bar{\Omega}_{i,j}$ is the time-averaged latency over interval T . This variance critically affects scheduling decisions and SLAs. Additionally, latency-sensitive applications such as real-time analytics and collaborative workloads impose latency upper-bound constraints, i.e.,

$$\mathbb{P}(T_u > \tau) \leq \epsilon \quad (32)$$

For some tolerance ϵ and threshold τ . This probabilistic constraint leads to the use of Chernoff bounds and tail-distribution modeling. Assume latency $T_u \sim \mathcal{N}(\mu_T, \sigma_T^2)$; then:

$$\mathbb{P}(T_u > \tau) \leq \exp\left(-\frac{(\tau - \mu_T)^2}{2\sigma_T^2}\right) \quad (33)$$

This constraint is embedded in optimization formulations as a soft reliability term. Network congestion can be modeled via queueing theory, using M/M/1 or M/G/1 queues. If arrival rate is λ , service rate is μ , then expected queueing delay is:

$$\mathbb{E}[L^{\text{queue}}] = \frac{\rho}{\mu(1 - \rho)}, \quad \text{where } \rho = \frac{\lambda}{\mu} \quad (34)$$

Such models guide load balancing and traffic shaping. For application-aware latency optimization, define a latency-aware utility function for data placement:

$$U(d_i) = \frac{1}{1 + \mathbb{E}[T_{d_i}]} \quad (35)$$

Which is maximized in the placement objective. Finally, in SDN-enabled hybrid clouds, dynamic latency maps are generated using telemetry $\mathcal{T}(t)$, allowing control-plane decisions based on:

$$\phi^*(d_i) = \arg \min_{n_j} \left(\mathbb{E}[T_{d_i}(t)] + \lambda_j(t) \cdot c_j \right) \quad (36)$$

where $\lambda_j(t)$ is the current load and c_j is the access cost. Thus, comprehensive latency and network analysis across hybrid and multi-cloud environments requires a confluence of queueing theory, stochastic modeling, dynamic graph algorithms, and real-time optimization—establishing the mathematical foundation for latency-sensitive orchestration, ensuring SLAs, and enabling high-performance distributed computing infrastructures.

2.7. Interoperability and service composition

In the context of multi-cloud and hybrid cloud environments, interoperability and service composition represent critical challenges due

to the inherent heterogeneity of platforms, data formats, APIs, and orchestration mechanisms (Shrivastava and Agrawal, 2024; Yeboah-Ofori et al., 2024). Interoperability, at its core, is the ability of systems and services from distinct cloud providers to exchange information and execute coordinated operations seamlessly (Pozveh et al., 2024). Let us define a multi-cloud system as a tuple $S = (C, I, P)$, where $C = \{C_1, C_2, \dots, C_m\}$ is the set of cloud providers, I is the set of interoperability mappings, and P denotes the pool of deployable services. Each service $S_i \in P$ is associated with an interface vector $\bar{\theta}_i \in \mathbb{R}^k$, encoding data schema, protocol stack, access control policies, and execution environment. The degree of interoperability between two services S_i and S_j can be mathematically modeled as a compatibility function $\delta(S_i, S_j)$ defined by:

$$\delta(S_i, S_j) = \exp\left(-\|\bar{\theta}_i - \bar{\theta}_j\|^2\right) \quad (37)$$

Which ranges in $[0, 1]$, with 1 denoting perfect compatibility. Services can only be composed if $\delta(S_i, S_j) \geq \tau$, where $\tau \in (0, 1)$ is a composition threshold. Now consider a service workflow $W = \{S_1, S_2, \dots, S_n\}$ across providers. The composition feasibility can be modeled via a service graph $G_W = (V, E)$, where each vertex $v_i \in V$ maps to a service S_i , and edges $e_{ij} \in E$ exist only if $\delta(S_i, S_j) \geq \tau$. The composability score of a workflow is defined as:

$$\Gamma(W) = \prod_{e_{ij} \in E} \delta(S_i, S_j) \quad (38)$$

Which penalizes weakly compatible links. When orchestrating such workflows, the optimization objective often involves minimizing the cumulative interoperability mismatch and latency across services:

$$\min_{W \subseteq P} \left(\sum_{e_{ij} \in E} (1 - \delta(S_i, S_j)) + \lambda \cdot \sum_{e_{ij} \in E} L_{ij} \right) \quad (39)$$

where λ is a trade-off weight balancing compatibility and latency. Furthermore, semantic heterogeneity is captured using ontology embeddings $\phi_i \in \mathbb{R}^d$ for each service S_i , and the semantic alignment between services is modeled as $\rho(S_i, S_j) = \phi_i^\top \phi_j$. In this setting, orchestration becomes a multi-objective problem involving structural, syntactic, and semantic dimensions. Consider a cross-platform service composition engine represented by a finite-state machine $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$, where Σ encodes service interface events and $\delta : Q \times \Sigma \rightarrow Q$ defines transition semantics; interoperability failure corresponds to the presence of absorbing error states $q_e \in Q$. The reliability of successful orchestration over k services can be expressed using a product of compatibility probabilities:

$$P_{\text{interop}} = \prod_{i=1}^{k-1} \delta(S_i, S_{i+1}) \quad (40)$$

Which quantifies the expected success rate of composing a k -length service chain. Hence, ensuring effective interoperability and service composition in multi-cloud ecosystems demands rigorous modeling using vector-space compatibility functions, graph-theoretic composition scoring, semantic embeddings, and automata-theoretic orchestration guarantees—ultimately translating the high-level design problem into a structured mathematical optimization framework over a heterogeneous and dynamic service landscape.

In addition, (Table 1) compares security levels across public, private, hybrid, and community clouds, evaluating confidentiality, integrity, availability, and compliance. Hybrid and private clouds excel due to advanced cryptography and access control, while public clouds struggle with confidentiality and key management. Multi-cloud security benefits from blockchain proofs, secret sharing, and federated identity, aligning with prior studies on structured cloud security frameworks.

The classification of security features in (Table 1), is based on a comprehensive analysis of the inherent characteristics and typical implementations of each cloud deployment model. Public clouds are generally designed with high availability due to their distributed nature

and redundancy across multiple data centers, which is why “Availability” is rated as high. Private clouds, while offering high availability, achieve this through dedicated resources and controlled environments, ensuring consistent uptime and reliability. Hybrid clouds combine the strengths of both, leveraging public cloud scalability and private cloud control, resulting in high availability as well. The ratings reflect industry standards and best practices, where these deployment models are known to prioritize availability through various mechanisms like load balancing, redundancy, and failover systems.

The justification for rating “Availability” as high in private clouds stems from their architecture, which is tailored to meet specific organizational needs with dedicated infrastructure. Unlike public clouds, where resources are shared, private clouds eliminate multi-tenancy risks, allowing for optimized performance and minimal downtime. Enterprises often implement advanced monitoring, disaster recovery plans, and localized redundancy in private clouds, ensuring high availability. For example, financial institutions or healthcare providers using private clouds can guarantee near-100% uptime due to their controlled environments and stringent SLAs. This aligns with documented case studies and cloud service benchmarks, where private clouds consistently demonstrate superior availability for critical applications.

The information in (Table 1), is derived from a synthesis of academic research and supported by practical implementations of cloud security frameworks (Chauhan and Shiales, 2023; Chang et al., 2016; Awayshah et al., 2021; Akinade et al., 2025; Osinuga et al., 2025; Hoz et al., 2025; Nwatuze et al., 2025; Nazir et al., 2025). Authoritative sources, such as NIST guidelines (Veeraballi, 2025; Bharathan, 2025; Dasmen et al., 2025; Chatterjee, 2021; Veeraballi, 2025), and studies on cloud deployment models (Mannan Javed et al., 2025; Patel and Kansara, 2021; Karnwal et al., 2011; Savu, 2011; Islam, 2024), were consulted to ensure accuracy. For instance, NIST SP 800-145 underscores the reliability of private clouds for mission-critical operations, while documentation from AWS and Azure highlights the redundancy measures inherent in public and hybrid models. Moreover, our analytical classifications are further validated by real-world examples—for example, private clouds employed by government agencies to achieve high availability through isolated and resilient infrastructures. Therefore, the ratings presented are not arbitrary but are grounded in empirical evidence and established principles of cloud security.

3. Security and privacy challenges in multi-cloud and hybrid cloud environments

The security and privacy challenges in multi-cloud and hybrid cloud environments involve numerous intricate issues that arise from the use of diverse cloud services in conjunction with on-premises infrastructure (Oladosu et al., 2021; Saxena et al., 2021). While these environments offer significant benefits in terms of flexibility, scalability, and cost efficiency, they also present serious security and privacy concerns that must be effectively addressed to meet regulatory standards and ensure robust protection. One of the most critical challenges is data security, encompassing data confidentiality, integrity, fragmentation, and the distribution of data storage, along with the secure management of data shares and encryption keys (Ali et al., 2025a, 2023). The protection of sensitive data integrity and confidentiality stands as an essential priority because various security models operate on heterogeneous platforms where information is processed (Islam, 2024; Sadique et al., 2025). When cloud service providers implement control methods for different distributed data sections the creation of security vulnerabilities becomes inevitable due to the diversity of protection mechanisms. Data security between cloud environments depends on proper encryption protocols for key management since any breach poses a serious threat to privacy. Structure-as-a-Service needs strong key encryption as well as secure storage systems and limited access rights alongside regular key replacement to protect against key vulnerabilities. All data-sharing operations need to have direct access controls that let authorized users or approved applications view and edit shared data.

Table 1
Comparison of security mechanisms across different cloud deployment models.

Security feature	Public cloud	Private cloud	Hybrid cloud	Community cloud	Authentication	Encryption	Access control	Attack resistance
Identity management	Moderate	High	High	Moderate	MFA, OAuth	AES-256, RSA	Role-based, attribute-based	Phishing, Spoofing protection
Confidentiality	Low	High	High	Moderate	Zero-knowledge proofs	Homomorphic, AES-256	Policy-based access	Insider threats protection
Data integrity	Moderate	High	High	High	Blockchain-based proofs	Hashing (SHA-256)	Smart contracts	Man-in-the-Middle attack protection
Availability	High	High	High	Moderate	Continuous monitoring	Redundancy	Load balancing	DDoS mitigation
Scalability	High	Moderate	High	Moderate	Federated identity	Dynamic key exchange	Policy-based expansion	Adaptive attack protection
Key management	Low	High	High	High	Hardware Security Modules (HSM)	Quantum-resistant algorithms	Distributed key storage	Secure multi-party computation
Secure data sharing	Low	High	High	High	Threshold cryptography	Secret sharing (Shamir's, CRT)	Fine-grained policy control	Secure multi-user access
Resilience	Low	High	High	Moderate	Self-healing systems	Decentralized storage	Adaptive AI models	Byzantine fault tolerance
Compliance	Moderate	High	High	High	GDPR, HIPAA	Encrypted logging	Fine-grained access	Legal & Regulatory adherence

3.1. Identity and Access Management (IAM) issues

Identity and access management (IAM) constitutes the primary organizational challenge for multi-cloud and hybrid cloud environments since it enables secure endpoint access to various platforms (Nair et al., 2024). Identity management across several cloud providers demands IAM systems that deliver uniform identity services between multiple providers who maintain separate authentication protocols and authorization policies (Badirova et al., 2023). Organizational complexity makes the selection process more challenging when organizations need to decide between role-based access control (RBAC) and attribute-based access control (ABAC). RBAC authorizes users by employing pre-defined roles yet does not provide sufficient precision for managing advanced environments because of their dynamically changing structure. Both ABAC and role-based access control utilize different approaches to access control where ABAC makes access decisions through attributes but manages them more complexly according to user location and device type with access time considerations (Servos and Osborn, 2017). A protective cloud asset environment depends on developing a single IAM framework that combines with hybrid and multi-cloud systems for enforcing access restriction and maintaining compliance standards.

3.2. Network security

Seamless communication networks represent a fundamental challenge in multi-cloud and hybrid cloud deployments due to unresolved security issues between interconnected cloud environments and the need to secure APIs and service invocation mechanisms (Yu et al., 2019). When data moves between cloud platforms through inter-cloud communication such systems expose confidential information to harmful interception attacks and potential tampering events during the transfer. The security threats can be reduced by implementing transport layer security (TLS) encryption protocols for achieving network data confidentiality and integrity (Parmar and Gosai, 2015). Internally and externally crucial APIs deliver security vulnerabilities because their protection status determines the level of threat they present. Unprotected APIs create easy entry points for unauthorized people who want to access services and data thus making them an enticing target for attackers. The protection of APIs requires multiple security measures that should include authentication solutions alongside encryption protocols and rate-limiting capability to stop exploitation according to both Kaul and Khurana (2021) and Qazi (2022). IT administrators need to implement stern rules that govern service-to-service communication between

cloud systems to stop unauthorized service contact thus decreasing access and disruptive risks.

3.3. Threats and attack vectors

Research studies analyze deliberate cloud-targeting techniques that affect operations that combine multiple clouds and hybrid clouds (Gupta et al., 2022b). One of the most prominent threats is the man-in-the-middle (MITM) attack, where an attacker intercepts communication between cloud services or between users and cloud platforms to extract sensitive information or inject malicious data (Tyagi et al., 2023). Additionally, side-channel attacks, which exploit physical or implementation-level vulnerabilities—such as timing information or power consumption—pose significant risks to the confidentiality of cloud-stored data (Inayat, 2017).

In multi-cloud settings, attackers may exploit security inconsistencies between different cloud providers, launching lateral movement attacks to escalate privileges and gain access across interconnected platforms. This cross-platform attack surface increases the likelihood of sophisticated breaches. A rapidly growing threat is ransomware, in which cybercriminals encrypt data stored in cloud environments and demand a ransom for decryption, posing severe operational and financial consequences (Ryan, 2021; Humayun et al., 2021). Another critical concern is data exfiltration, where attackers stealthily extract data from cloud platforms. This becomes particularly problematic in environments where data is distributed across multiple providers and geographic locations (Ullah et al., 2018).

These emerging threats show that organizations are at a high risk of facing new and complex threats, hence the need to adopt robust and matrix defense strategies. Measures range from constant vigilance, network Security Information and Event Management, and the proper intrusion detection and prevention systems (IDPS) for advanced and upcoming threats that may be hazardous in today's world.

3.4. Regulatory and compliance challenges

Compliance and regulatory challenges significantly compound the issues of security and privacy in multi-cloud and hybrid cloud environments (Julakanti et al., 2022). The rapid advancement of cloud technologies has introduced numerous regulatory obligations for organizations, including the general data protection regulation (GDPR) in the European Union, the california consumer privacy act (CCPA) in the United States, and data localization laws that mandate local storage

and processing of data (Mbah, 2024; Voss, 2019). Strict adherence to these regulations is essential to avoid legal consequences and preserve consumer trust. However, achieving compliance becomes increasingly complex when leveraging multiple cloud service providers that operate across various jurisdictions, each with distinct legal requirements.

The two main concerns that are relevant to the current problem are the legal aspect of the issue and trust jurisdiction. To address this challenge organizations must ensure that the providers are required to be legal and ought to be able to face the law in the countries of service delivery to prosecute them for violation of the laws of the country. This requires great caution when choosing a provider to work with; additional emphasis should be placed on having proper legal compliance and having the input legal contracts in place to outline responsibilities and liabilities dating between the providers and the organization.

This subsection also discusses the numerous security and privacy concerns about the multi-cloud and hybrid cloud environments, hence calling for a detailed approach that integrates into the security strategy. Issues like sensitivity, distribution, fragmentation, confidentiality, and integrity of data, as well as the proper way of secure sharing of data, have to be deliberately treated and solved so as not to make the information get into the wrong hands and be leaked. Hence, features such as cross-cloud authentication and implementable models like RBAC and ABAC should be made more precise so that restrictive access control is achieved (Gupta et al., 2022a).

However, the security aspect is still an issue because information interchange between the clouds and during the API interactions has to be passed through secure channels and constantly monitored to avoid the risks of potential MitM attacks (Akhtar et al., 2022). Other persistent and evolving threats—including side-channel attacks, ransomware, and data exfiltration—require dynamic and adaptive security mechanisms to ensure proactive defense.

Lastly, regulatory constraints such as GDPR, CCPA, and data sovereignty laws present intricate legal challenges that influence how organizations interact with cloud providers (Somasundaram, 2022; Fischer, 2023). Addressing these challenges necessitates a multi-pronged approach, combining technological innovation, policy development, and regulatory alignment to enable secure utilization of multi-cloud and hybrid cloud architectures without compromising organizational data integrity or eroding customer and stakeholder trust.

The (Table 2) presents a comparative summary of the security and privacy issues associated with multi-cloud and hybrid cloud environments, highlighting their implications and potential mitigation strategies. Security remains a critical concern, as applications continue to face challenges in ensuring data confidentiality and integrity, managing cryptographic keys, and implementing efficient and secure key distribution mechanisms.

The implementation of identity and access management (IAM) creates specific hurdles that become considerable when dealing with cross-cloud identity management alongside attribute-based access control and the choice between role-based access control (RBAC) and attribute-based access control (ABAC). To ensure consistent and secure platform access across varied cloud environments organizations need to use federated identity models combined with context-aware access control frameworks.

In addition, inter-cloud communication requires the use of strong encryption measures and where necessary use of tokens to reduce the risks that characterize transfer of data and sharing in different clouds.

Compliance with regulatory requirements such as the general data protection regulation (GDPR) and the California consumer privacy act (CCPA) further complicates these concerns. Specifically, issues related to data localization, jurisdictional conflicts, and the enforcement of legal obligations must be addressed through automation, well-defined service-level agreements (SLAs), and policy-aware cloud service orchestration.

4. Limitations of multi-cloud approach and practical implementation suggestions

Limitation 1: Personal Data Storage Requirements (Data Residency Laws:)

Data residency laws refer to legal requirements that personal data must be stored and processed within a specific geographic location—usually the country or region where the individual resides. This ensures national control over citizens' sensitive information and prevents unauthorized foreign surveillance or exploitation.

In a multi-cloud architecture, organizations use services from multiple cloud vendors (e.g., AWS, Azure, GCP, Alibaba Cloud). These vendors have global infrastructures, often routing or replicating data across different data centers for redundancy, latency optimization, or cost efficiency.

Problem: This automated routing may inadvertently result in:

- Data being stored or backed up in foreign regions, violating data localization laws.
- Processing occurring outside national borders, exposing data to foreign jurisdictions.

Example: Suppose a healthcare provider in Germany stores patient data on AWS, while also using Azure AI services for analytics. Even if both vendors comply with GDPR unless explicit regional controls are configured, the analytics data might be processed in Ireland or the US, a potential violation of GDPR Article 44 on international data transfer.

Limitation 2: Compliance Challenges in Multi-Cloud Scenarios:

Even if cloud providers claim regulatory compliance, the ultimate legal responsibility for data protection lies with the data controller, the organization that uses the cloud. Different clouds offer different tools, interfaces, and policies, creating fragmentation in compliance management.

Specific Challenges follow as:

1. lack of visibility and control—organizations often struggle to track where data is stored or replicated, increasing the risk of non-compliance with data retention policies.
2. Complex data governance further complicates matters, as each cloud provider employs distinct identity and access management (IAM) models, varying encryption standards, and key management options, while often lacking centralized policy enforcement across platforms.
3. Inconsistent logging and auditing pose significant hurdles, with disparate tools like AWS CloudTrail, Azure Monitor, and Google Cloud Audit Logs making it difficult to correlate logs for unified reporting, forcing organizations to rely on error-prone manual aggregation that lacks real-time detection capabilities.
4. cross-border data flows introduce legal and compliance risks, as APIs, CDNs, and machine learning models may inadvertently transfer data across jurisdictions, potentially violating geographic regulations—especially during dynamic processes like load balancing, where users might unknowingly trigger such breaches.

Together, these challenges create a fragmented security and compliance landscape, demanding robust solutions for centralized oversight and policy harmonization.

Solution 1: Data Classification and Localization Policies:

To specify both data sensitivity levels and storage area compliance organizations need to implement tagging protocols during data classification. Data storage is limited to particular geographic locations under the obligations of localization requirements.

The interoperative usage of Azure Information Protection tools alongside AWS Macie and Google Cloud Data Loss Prevention allows organizations to detect personal and financial records across multiple cloud platforms for classification initiatives. The tools support predefined data handling categories for maintaining the constant safety of sensitive assets. A healthcare center needs an automation system for metadata tagging which follows their organization policies by applying

Table 2

Comparison of security and privacy challenges in multi-cloud and hybrid cloud environments.

Category	Challenges	Implications	Mitigation strategies
Data security	Data confidentiality and integrity	Risk of unauthorized access, tampering, and data breaches	Strong encryption (AES-256, Homomorphic), integrity checks (hashing)
	Data fragmentation and storage distribution	Complexity in security enforcement across different cloud providers	Data classification, unified security policies, distributed access control
	Secure data sharing and key management	Risk of unauthorized access due to weak key management	Secure Multi-Party Computation (SMPC), Hardware Security Modules (HSM), Threshold cryptography
Identity and Access Management (IAM)	Cross-cloud authentication and authorization	Difficulty in seamless integration of IAM across cloud providers	Federated identity, Single Sign-On (SSO), Multi-Factor Authentication (MFA)
	RBAC vs. ABAC selection	RBAC lacks flexibility, ABAC is complex to manage	Hybrid IAM models combining RBAC and ABAC for adaptive access control
Network security	Inter-cloud communication security	Risk of data interception and tampering during cloud interactions	Transport Layer Security (TLS), End-to-End Encryption (E2EE)
	Secure API and service invocation	Vulnerability to API-based attacks	API gateways, token-based authentication (OAuth), rate limiting
Threats and attack vectors	Man-in-the-Middle (MitM) attacks	Unauthorized interception of communication	Strong encryption, secure key exchange protocols
	Side-channel attacks	Exploitation of hardware-based vulnerabilities	Secure hardware modules (TPMs), randomized execution strategies
	Cross-cloud attacks	Exploiting weak security in different clouds	Unified threat intelligence, consistent security policies across clouds
	Ransomware	Data encryption and ransom demands	Backup strategies, anomaly detection, ransomware-resistant file systems
	Data exfiltration	Unauthorized data leakage	Data Loss Prevention (DLP), real-time monitoring
Regulatory and compliance	Compliance with GDPR, CCPA, and data localization laws	Complexity in ensuring compliance across multiple jurisdictions	Compliance automation, cloud provider agreements, data residency controls
	Trust and legal jurisdiction issues	Uncertainty in regulatory enforcement and liability	Transparent SLAs, independent security audits, legal contracts

Sensitivity = High combined with Region = EU to data attributes. The system allows organizations to monitor operations continuously which helps maintain policy defensive requirements. Data transfer boundary restrictions require integrated geo-fencing methods which unite AWS location restriction policies with Azure's data residency constraints and Google Cloud's Regional Service Usage Policies. The exact access privileges of users who hold authorization permit them to view restricted data along with storage controls that stop unauthorized movements of information between regulated areas. A distributed cloud implementation with this approach allows organizations to achieve regulatory compliance in addition to improving data protection and entire system security.

Solution 2: Centralized Data Governance Framework:

A unified governance strategy allows central enforcement of privacy, access control, and retention policies across multiple clouds.

Code compliance rules require enforcement across multiple clouds through an OPA-open policy agent together with terraform and cloud custodian-based automation for policy governance. Multiple cloud platforms use these tools for users to develop common security and compliance rules that enable universal enforcement across their cloud environments. Departments should deploy SSO and IAM federation solutions because the solutions provide unified identity management systems to enable RBAC security across different cloud providers. Standardized policy enforcement through an integrated security structure ensures the security protocols remain consistent despite different vendor systems below them thus lowering the risk of policy inconsistencies and human errors. These programming tools allow organizations to achieve operational excellence through policy-based scalable controls in multiple cloud networks to secure their position.

Solution 3: Privacy Enhancing Technologies (PETs):

PETs reduce the exposure of sensitive data by processing them in a way that preserves privacy. Even if data crosses borders, they become useless to adversaries without decryption.

Privacy-enhancing technologies (PETs) provide cryptographic along statistical solutions to secure data processing operations within multi-cloud environments. The security feature of HE permits calculations on

encrypted files before encryption thus helping privacy-centric machine learning operations in diverse cloud environments. Joint data processing through secure multi-party computation (SMPC) allows multiple parties to jointly work with data but stops any entity from seeing the entire dataset while maintaining complete information confidentiality. The addition of controlled noise to result in outputs through Differential Privacy methods makes it impossible for any party to retrieve original sensitive data from the information released. Tokenization allows protected tokens to substitute important data elements so cloud processing will operate without divulging genuine data content since all original records remain safely stored in on-premise vaults. Data sovereignty compliance occurs through combined data privacy mechanisms which stop authorized geographic areas from losing personal or controlled data when processing in multiple cloud platforms thus protecting privacy without limiting international analytics or collaboration among teams.

Solution 4: Data Sovereignty-Aware Cloud Providers:

Some cloud providers offer sovereign clouds—cloud regions designed specifically to comply with national regulations, operated in cooperation with local governments or companies.

Major cloud service providers provide sovereign cloud solutions as national data protection and regulatory standards compliance solutions for their international markets. Organizations in the public sector maintain data residence through Microsoft Cloud for Sovereignty and AWS GovCloud which offer localized control systems and U.S. government regulatory compliance respectively. Thales Sovereign Cloud provides data storage services that follow the data protection norms of the general data protection regulation (GDPR) to support secure national storage of sensitive information. The sovereign cloud infrastructure offers multiple protective features which consist of data accreditation standards alongside government-sanctioned encryption key management procedures. Programs that let organizations customize their cloud workforce migrations enable them to follow national laws while reducing regulatory risks and maintaining complete data boundary control.

Solution 5: Third-Party Compliance Assurance:

Using external assessments and certifications helps ensure compliance is objectively validated across clouds.

Multi-cloud environments necessitate the involvement of independent auditors to assess GDPR compliance, particularly in relation to Article 25 (Data Protection by Design), Article 30 (Records of Processing Activities), and Article 32 (Security of Processing). To achieve global certification compliance, organizations must implement security platforms that integrate standards such as ISO/IEC 27001 for information security management, SOC 2 Type II for operational controls, and CSA STAR Level 2 for cloud-specific assurance. Leveraging automated compliance workflow systems—such as OneTrust, TrustArc, and IBM OpenPages—enables organizations to improve the efficiency and accuracy of risk assessments and audit reporting by minimizing documentation burdens and reducing the likelihood of operational errors. Organizations that adopt a structured compliance strategy gain the ability to track regulatory updates, satisfy legal obligations, and foster stronger stakeholder trust through proactive data governance. In distributed cloud architectures, maintaining operational efficiency depends on a combination of manual and automated compliance management tools, independent third-party audits, and adherence to recognized industry certification frameworks.

4.1. Interoperability challenges across multi-cloud and hybrid cloud environments

4.1.1. Federated identity management across clouds

To resolve the interoperability issues in identity and access management (IAM) across heterogeneous cloud services (Indu et al., 2018; Esposito, 2018), it is essential to adopt a federated identity model in which a central identity provider (IdP) handles authentication and authorization across all cloud platforms. This can be implemented using widely accepted identity federation protocols such as SAML 2.0, OAuth 2.0, and OpenID Connect (Naik and Jenkins, 2017), which allow users authenticated in one domain to seamlessly access resources in another without re-authentication. By integrating the native IAM of each cloud provider (e.g., AWS IAM, Azure Active Directory, and Google Cloud IAM) with a centralized IdP (Di Federico and Barcaroli, 2022), a uniform trust mechanism can be established by which token-based assertions or claims from the IdP are recognized and honored in all cloud services. A unified security structure is promoted because users no longer need multiple copies of their identity or a fragmented approach when enforcing security policies. Safety practices require organizations to select standardized policy definitions and provisioning protocols such as XACML and SCIM to define user characteristics roles and permissions which all participating systems can read in a standardized format. An identity translation gateway functions as an intermediary system that resolves protocol translation difficulties by converting JWT-based tokens from one system to SAML assertions for another platform during the process of user attribute and access scope and group membership mapping across multiple cloud platform schema definitions. The gateway provides a mechanism that allows trouble-free identity transport between different trust relationships and uninterrupted access throughout different security domains.

4.1.2. Centralized governance and cloud-native policy normalization

Beyond identity federation, a unified governance framework must be established to centralize reporting, policy management, and compliance auditing in all clouds. This is achievable through cloud-agnostic cloud security posture management (CSPM) tools that continuously monitor IAM configurations, misconfigurations, and user activity logs (Baladari, 2024), then route them into a centralized security information and event management (SIEM) system such as ELK Stack, Splunk, or Azure Sentinel. These logs should be normalized using structured event schemas such as the CloudEvents standard to ensure semantic consistency in security alerts and forensic records regardless

of their origin cloud. This centralization enables real-time detection of anomalies, lateral movements, and policy violations while facilitating compliance reporting for frameworks like GDPR, HIPAA, or ISO/IEC 27001 (Diamantopoulou et al., 2020). In parallel, organizations should enforce a zero trust architecture (ZTA) that treats identity as the new perimeter and enforces continuous risk-based verification of users and devices. This involves calculating a dynamic trust score based on factors such as authentication strength (e.g. multifactor authentication), device health, geolocation, and behavioral consistency. Mathematically, if $T(u, r) = f(A, D, C, P)$ represents a trust function that evaluates the request of the user u to the resource r , access is only granted if $T(u, r) \geq \tau$, where τ is an adaptive risk threshold. Such probabilistic trust enforcement prevents unauthorized access even in the event of a credential compromise, as contextual signals must also align with expected behavior patterns.

4.1.3. Granular access control via ABAC and policy engines

The rigidity and difficulties with scaling apply to role-based access control (RBAC) in their traditional form when applied to active multi-cloud environments. ABAC yields a refined access control system because access decisions result from multiple evaluations between user attributes (e.g., department, security clearance) and resource properties (e.g. confidentiality level, owner tag) and environmental variables (e.g., access time, network location). When all conditions defined within the policy logic match the policies stored in the open policy agent (OPA) (Ferraiolo et al., 2011), access permission is automatically granted. Policies operate independently from fixed role mappings because they apply real-time evaluations that foster enhanced security together with better flexibility. The deployment of infrastructure-as-code (IaC) tools includes Terraform and Pulumi to provision IAM roles and add permissions and access policies and federation trusts declaratively, which results in reproducible auditable and version-controlled IAM infrastructure. The tools perform automated detection of drifts in the access control policy and automatically rectify these issues to preserve the integrity of the framework according to JohnPaul and Nwalozie (2024).

4.1.4. Formal verification and interoperability testing framework

The achievement of IAM interoperability requires thorough testing and simulation of cloud environments according to the findings in Akinade et al. (2025). Runtime failures of syntactically correct policies stem from semantic mismatch errors together with unexpected privilege-escalation paths. A formal test harness must be constructed to detect and address access-related problems by verifying policy compliance while replicating access requests throughout all connected clouds (Mazhar et al., 2025). Using formal verification methods with tools such as Alloy or TLA+ it becomes possible to verify the access control graph for detecting unauthorized security problems in the form of transitive access or role inheritance loops. The access control inheritance structure through IAM roles forms a directed graph $G = (V, E)$ with roles as nodes and delegation or inheritance as edges. A different set of available privileges permits us to see possible escalation routes through depth-first-search (DFS) procedures, which support pending enhancements to security policy structures before implementation. Interoperability among various IAM solutions in hybrid and multi-cloud platforms becomes achievable through identity federation together with standardized policy enforcement and trust-based access evaluation supporting automation formal verification and centralized governance.

5. Existing security and privacy strategies

These security and privacy strategies present extensive information about diverse techniques used against security and privacy threats that affect multi-cloud and hybrid-cloud systems. The security enhancement strategies include cryptographic methods together with identity management systems secure communication protocols and artificial

intelligence systems which maintain security requirements while adhering to legal framework regulations (Habbal et al., 2024; Aisyah et al., 2019; Villegas-Ch and García-Ortiz, 2023; Gbadebo et al., 2024; Zafir et al., 2024).

Secure storage and information transfer between different cloud services depend on robust data security measures and integrity (Gupta et al., 2022c; Kaaniche and Laurent, 2017; Yang et al., 2020; Sun, 2019), which are critical components of cryptographic solutions. Homomorphic encryption allows users to perform data processing directly on encrypted data, ensuring that encryption is not required beforehand. This security method preserves data confidentiality throughout the processing phase (Acar et al., 2018). High-security industries, such as healthcare and financial services, derive maximum benefits from these capabilities (Council et al., 1997).

Users can execute joint input data operations through secure multiparty computation (SMPC) without exposing or disclosing their source information. Users depend on this technology to run distributed decision-making workflows that handle analyses of joint data from multiple databases without exposing original information confidentiality.

The combination of blockchain technology with zero-knowledge proofs (ZKPs) ensures both data confidentiality and information integrity as per (Li et al., 2020b; Huang et al., 2022). The spreading nature of blockchain technology generates permanent records that verify that commands remain unmodified. ZKPs enable users to demonstrate valid information to others by protecting all other details from disclosure throughout transactions (Sun et al., 2021).

The cryptographic tools provide joint data security that enables protection for hybrid cloud and multi-cloud infrastructure systems.

This is a key factor because it enables effective control over various cloud resources by clearly defining access privileges across different cloud deployments and environments (Andrikopoulos et al., 2013; Manvi and Shyam, 2014). Multifactor authentication (MFA), a widely adopted identity and access management (IAM) solution, improves security by requiring users to present multiple forms of verification to gain access, thus reducing the risk of identity theft (Otta, 2023). Typically, MFA consists of three categories: something the user knows (for example, a password), something the user possesses (for example, a smartphone or hardware token), and something the user is (for example, biometric features such as fingerprints or facial recognition) (Shukla et al., 2024).

Federated identity management is another important IAM approach, allowing users to access multiple cloud services using a single set of credentials (Habiba et al., 2014). This not only simplifies the user experience but also improves security by reducing the number of entities that require credential input.

The emerging organizational concept called decentralized identity management bases its operation on blockchain technology. Users gain complete control of their digital identities under this management approach by doing away with centralized authorities. Users under this model can privately administer their identity information without needing government or organizational supervision. The method proves essential in multi-cloud operations since users work with multiple vendor services each operating different identity management platforms. Organizations that use decentralized identity management can better govern and control access to their multiple cloud platforms (Sung and Park, 2021; Dib and Rababah, 2020).

Secure communication and network defense mechanisms: As data flows between cloud environments and systems, including those between the cloud and the on-premises infrastructure of an organization, robust security measures must be implemented (Jansen et al., 2011). Software-defined perimeter (SDP) and zero trust architecture (ZTA) are two emerging approaches to network security that have gained recent attention. In particular, SDP establishes an adaptive security boundary around cloud resources, ensuring that only authenticated users or devices can access them. This approach is particularly valuable in multi-cloud configurations, where traditional perimeter-based security mechanisms may be inadequate.

ZTA, guided by the principle “never trust, always verify”, continuously authenticates users and assesses device integrity before granting access to any resource. This prevents unauthorized users from accessing restricted network segments and inhibits lateral movement within the network (Powell, 2020).

Various security tools such as VPN alongside SSL and TLS combine to provide users with protected transmission of data that maintains both confidentiality and integrity. Real-time traffic monitoring and threat analysis functions are provided by cloud-based intrusion detection systems (IDS) which serve as a vital security installation. These detection systems monitor for different forms of cyber threats such as unauthorized access attempts together with malware infections and denial-of-service attacks to provide organizations with a prior warning for effective counteraction (Snehi and Bhandari, 2021).

Artificial Intelligence (AI) and Machine Learning (ML): These two technological solutions actively boost threat detection processes and emergency response operations within hybrid and multi-cloud systems. Network patterns and abnormal behaviors enable the anomaly-based approach to detect security threats efficiently (Garcia-Teodoro et al., 2009). The combination of rapid detection and analytical functions allows this approach to identify security risks that human analysts may miss by detecting irregular system patterns to boost company defense capabilities.

The computerization of threat intelligence sharing enabled by AI serves as a strategic approach for the real-time distribution of new threat information and security vulnerabilities. By operating in unison security of cloud environments improves since stakeholder members join together to detect threats and prevent upcoming security risks.

Risk analysis prediction utilizes artificial intelligence (AI) and machine learning (ML) as components for its performance. Through historical data analysis, these technologies detect criminal patterns for establishing crime predictions which help organizations avoid safety incidents (Shah et al., 2021).

Privacy-enhancing technologies: These measures are crucial to ensure that data stored in the cloud remains inaccessible to unauthorized users or tenants. Differential privacy is a widely adopted method that introduces noise to datasets, ensuring that individual records cannot be uniquely identified while preserving sufficient utility for analysis and inference (Zhu et al., 2017; Ji et al., 2014; Hassan et al., 2019; Zhao et al., 2019). This technique is particularly effective in environments that require data sharing or analysis between organizations, such as healthcare or finance, where data sensitivity is paramount.

Federated learning provides organic entities with a method to train machine learning models collectively without putting their actual data at risk. Each party keeps their data private while performing local training and they share only the model's updates instead of actual data through this approach. Advanced federated learning systems merge homomorphic encryption and secure multi-party computation (SMPC) as their main components for protecting training process security and privacy (Xie et al., 2024; Kanagavelu et al., 2020). These security methods combine to provide organizations a strong approach for maintaining confidentiality while leveraging cloud infrastructure across multi-cloud systems and hybrid cloud environments.

Finally, the section Current Security and Privacy Trends highlights the emergence of sophisticated mechanisms addressing security and privacy challenges in multi-cloud and hybrid-cloud environments. Techniques such as homomorphic encryption, multi-party secure computation, blockchain, and zero-knowledge proofs form the foundational pillars for privacy-preserving data storage and computation (Xu et al., 2024; Bernabe et al., 2019; Wang et al., 2020b). Identity and access management (IAM) mechanisms—including multifactor authentication, federated identity management, and decentralized identity management—ensure that only authorized users and devices gain access to cloud resources. To protect data in transit across cloud environments, network-level protections such as software-defined perimeter

Table 3

Comparison of existing security and privacy strategies in multi-Cloud and hybrid cloud environments.

Security & Privacy strategy	Key features	Advantages	Challenges
Cryptographic techniques	Homomorphic encryption, Secure Multi-Party Computation (SMPC), Blockchain, Zero-Knowledge Proofs (ZKP)	Ensures confidentiality and integrity, enables secure computations on encrypted data, provides immutability and trust	High computational overhead, integration complexity, limited scalability in large-scale multi-cloud deployments
Identity and Access Management (IAM)	Multi-Factor Authentication (MFA), Federated identity, Decentralized identity	Strong user authentication, minimizes password-based risks, enhances user control over identity attributes	Integration challenges across diverse platforms, risk of single point of failure (Federated IAM), slow adoption of decentralized identity
Secure communication & Network defense	Software-Defined Perimeter (SDP), Zero Trust Architecture (ZTA), VPNs, SSL/TLS, Cloud-based IDS	Secures data-in-transit, implements least privilege access, enables real-time intrusion detection	Network latency, deployment complexity, risk of misconfiguration, false positives in IDS
AI & Machine learning for threat detection	AI-based anomaly detection, Threat intelligence sharing, Predictive analytics	Enables proactive threat detection, real-time response, enhanced attack surface visibility	Requires large datasets, prone to adversarial manipulation, high false positive rates
Privacy-Preserving techniques	Differential privacy, Federated learning, Encrypted model training	Enables secure AI training without central data pooling, enhances user privacy	Accuracy-privacy trade-offs, higher compute costs, regulatory complexities across jurisdictions

(SDP), zero trust architecture (ZTA), encrypted tunneling, and cloud-based intrusion detection systems (IDS) are employed.

Artificial intelligence (AI) techniques—such as anomaly detection, automated threat intelligence sharing, and predictive analytics—play a pivotal role in improving threat detection and incident response (Aminu et al., 2024; Ekundayo et al., 2024; Bécue et al., 2021). Moreover, privacy-preserving approaches such as differential privacy, federated learning, and secure model training ensure data security even in multi-tenant and data-sharing scenarios (Zhang et al., 2021a; Yang et al., 2019; Li et al., 2020a). Altogether, these strategies empower organizations to manage the complexity of multi-cloud and hybrid cloud environments effectively, ensuring data security and maintaining the trust of clients and stakeholders.

The existing security and privacy strategies used in multi-cloud and hybrid cloud environments are summarized in (Table 3). Encryption techniques that ensure data confidentiality—such as homomorphic encryption and secure multiparty computation—are effective, but often come with significant computational overhead. Identity and access management mechanisms, including multifactor authentication and federated identity management, also provide robust security; however, they can suffer from interoperability and interconnectivity challenges across different platforms.

Emerging approaches for secure data transmission, such as zero trust architecture (ZTA) and software-defined perimeter (SDP), offer enhanced protection by minimizing trust assumptions and enforcing strict access controls. Despite their security benefits, both ZTA and SDP present substantial implementation complexities.

Furthermore, artificial intelligence (AI) has contributed significantly to the advancement of threat detection and privacy-preserving mechanisms. However, concerns remain regarding the vulnerability of AI models to adversarial attacks, as well as the high computational costs associated with deploying such intelligent systems at scale.

6. Comparative analysis of security strategies

The research explores security, efficiency, and usability aspects of emerging multi-cloud and hybrid cloud environments. A comparative analysis of security operation costs is included in the study, along with multiple trade-offs encountered when managing multi-cloud environments. Actual implementation examples are provided to illustrate the current maturity of multi-cloud architectures. Numerous security solutions have been developed in response to regulatory requirements for data protection; however, these solutions can negatively impact operational speed, thereby affecting user experience and response times (Habibzadeh et al., 2019; Bergkamp, 2002). Homomorphic encryption and secure multi-party computation (SMPC) require extensive computational power which means organizations need to use substantial system resources according to studies in rahaman2024secure and zhou2024secure. Real-time data integrity checks

and immutability with blockchain or distributed ledger technologies have substantial requirements for computational power together with storage capacity and system resources (Habib et al., 2022; Reyna et al., 2018). Approximately large datasets cause the computational expenses of zero-knowledge proofs (ZKPs) to rise significantly (Lavin et al., 2024), while these proofs maintain robust privacy security. To obtain strong security organizations must balance their needs for system performance against usability requirements while finding safe and convenient solutions that fit their requirements.

Data and service distribution across multiple geographic regions make management challenges grow more important in multi-cloud and hybrid cloud setups (Merseedi and Zeebaree, 2024). Through identity and access management (IAM) solutions organizations gain security benefits because MFA and federated identity management let them control access to designated resources by authorized users. The implementation of security protocols creates usability challenges because users must manage many identity items across diverse network systems using complicated authentication methods. Decentralized identity management stands apart from centralized approaches in privacy terms but requires significant investment in infrastructure together with experienced IT personnel to sustain its operation. Both SDP and ZTA security protocols lead to decreased system performance although they effectively protect networks against threats. VPNs and SSL/TLS operate as secure tunneling methods though users experience latency in transmission and need substantial setting and upkeep (Abbas et al., 2023; Gaur et al., 2021). Cloud-based intrusion detection system (IDS) services provide real-time threat detection and surveillance while successfully monitoring networks but result in numerous alerts that might exceed system response capacity.

Examples of security implementation practiced by organizations will be explored in the upcoming section for their multi-cloud and hybrid architectural configurations. The research examples show business organizations as they balance security needs with network speed requirements and end-user comfort expectations. Healthcare providers use homomorphic encryption integrated with federated learning to obtain safe information-sharing capabilities along with analysis across multiple platforms. Security in financial institutions becomes stronger through their combination of zero-trust architecture and AI-based anomaly detection to protect their network from threats. The examined deployment examples analyze the practical implementation challenges of cloud security technologies for different organizational sectors. They show how security solutions work with various industries.

A detailed summary of the influence of security mechanisms on performance characteristics, including computational overhead, is presented in Table 4 to support the comparative analysis.

The implementation of multi-cloud and hybrid cloud solutions demands security effectiveness evaluations combined with expense assessments in security assessments. The security assessment method helps organizations achieve proper security-performance-comfort trades for

Table 4
Security mechanisms, performance overhead, and trade-offs.

Security mechanism	Performance and overhead	Trade-offs
Homomorphic encryption	High computational cost, slow processing times, increased memory and CPU usage	Strong data confidentiality on encrypted data but reduced system throughput
Secure Multi-Party Computation (SMPC)	Computationally intensive with significant network and processing overhead	Robust data privacy in distributed systems, but complex protocol design and resource demands
Blockchain technology	Resource-heavy consensus mechanisms, high latency, and substantial storage needs	Immutability and auditability of records, but limited scalability and operational efficiency
Zero-Knowledge Proofs (ZKPs)	Expensive in terms of computation and memory, particularly for complex inputs	Strong privacy without revealing sensitive data, at the cost of performance overhead
Multi-Factor Authentication (MFA)	Requires additional steps and infrastructure for user validation	Increased login time and possible user friction, but strong resistance to credential theft
Federated identity management	Interoperability complexity and management overhead across platforms	Simplified user experience across clouds, but higher integration and administrative burden
Decentralized Identity (DID)	Requires blockchain or similar infrastructure, with steep learning curve	Improved user control and privacy, but adoption and implementation challenges
Software-Defined Perimeter (SDP)	Adds configuration complexity and latency due to dynamic perimeters	Strong micro-segmentation and access control, but may reduce user transparency and speed
Zero Trust Architecture (ZTA)	Continuous verification incurs latency and requires constant monitoring	High security assurance but potential for increased processing load and delay
Secure Tunneling (VPNs, SSL/TLS)	Introduces overhead due to encryption and tunneling protocols	Ensures secure data-in-transit but may reduce throughput and increase latency
Cloud-Based Intrusion Detection Systems (IDS)	Generates voluminous alerts; requires analytics and response infrastructure	Real-time attack detection but resource-intensive with potential for alert fatigue
AI-Driven anomaly detection	Demands high-performance computing and model training resources	Detects novel threats effectively, but susceptible to false positives and adversarial data
Automated threat intelligence sharing	Needs robust interoperability and secure channels; risk of redundant data	Faster, coordinated threat responses but can overwhelm analysts with excess input
Differential privacy	Adds statistical noise to results, impacting output fidelity	Maintains user anonymity but may degrade data utility and model accuracy
Federated learning	Requires model synchronization and communication among nodes	Preserves local data privacy but introduces communication and training delays
Secure model training	Leverages encryption and privacy protocols, increasing computation load	Protects sensitive training data but leads to slower convergence and energy costs

their systems. Real-world examples mentioned in this paper enable organizations to create workable security plans that maintain maximum system performance for better user satisfaction. The table attached furnishes detailed information that describes different security methods together with their benefits and weaknesses. This reference serves organizations seeking to establish thorough and secure distributed cloud environment security strategies. Organizations can establish and refine security plans before the arrival of multiple cloud issues through implementation and evaluation so they can defend sensitive data effectively while building customer trust.

The (Fig. 3), presents a comparative analysis of security and privacy metrics between Multi-Cloud and Hybrid Cloud environments, where performance scores (ranging from 0 to 100) are plotted across various key parameters. The Multi-Cloud model, represented by a solid blue line with circular markers, generally outperforms the Hybrid Cloud, depicted by a red dashed line with square markers, across most metrics, except for Integrity and Attack Resilience, where the Hybrid Cloud achieves slightly higher scores. Both models peak in availability, with Multi-Cloud showing a marginal lead, while latency demonstrates a noticeable dip for both, with Hybrid Cloud performing slightly worse. Throughput demonstrates better performance in Hybrid Cloud environments, while computational overhead remains generally lower, indicating greater performance efficiency. The persistence of attacks suggests that Hybrid Cloud implementations offer robust security benefits across their domains. The analysis shows that Multi-Cloud excels in availability and confidentiality, whereas Hybrid Cloud exhibits greater resilience to threats and reduced resource utilization. Performance indicators across different cloud environments reveal that security and efficiency often trade off directly with reliability in cloud computing infrastructures. These research conclusions offer valuable

guidance to organizations in selecting cloud service architectures that align with both their security requirements and operational demands.

7. Emerging trends and future directions

The discussion centers on future security trends in cyber defense within cloud environments, with particular emphasis on emerging directions aimed at building resilient digital infrastructures. A critical component of cloud security is the integration of post-quantum cryptography (PQC), as experts widely anticipate that current cryptographic schemes will become vulnerable in the face of quantum computing advancements. Quantum computers operating at full capacity are expected to compromise traditional encryption methods such as RSA and elliptic curve cryptography (ECC), along with various other algorithms presently in widespread use. The primary objective of PQC is to develop cryptographic algorithms that are inherently resistant to quantum attacks, thereby securing both cloud-based data storage and communication channels. The transition to post-quantum solutions is imperative for cloud security, given the vast volumes of sensitive information housed within cloud infrastructures, which renders them highly susceptible to emerging quantum-enabled threats.

Another significant trend is the rise of AI-Driven Self-Adaptive Security Mechanisms. As the name implies, these systems use artificial intelligence and machine learning to develop security mechanisms that are self-tuning and capable of autonomously detecting and responding to threats in real-time. AI-driven solutions can analyze massive volumes of data, identify anomalous behavior or threat patterns, and dynamically update defensive strategies. This adaptability allows cloud systems to respond to new and evolving cyber threats more effectively than traditional rule-based methods, which often fail to keep up with the sophistication of modern attacks.

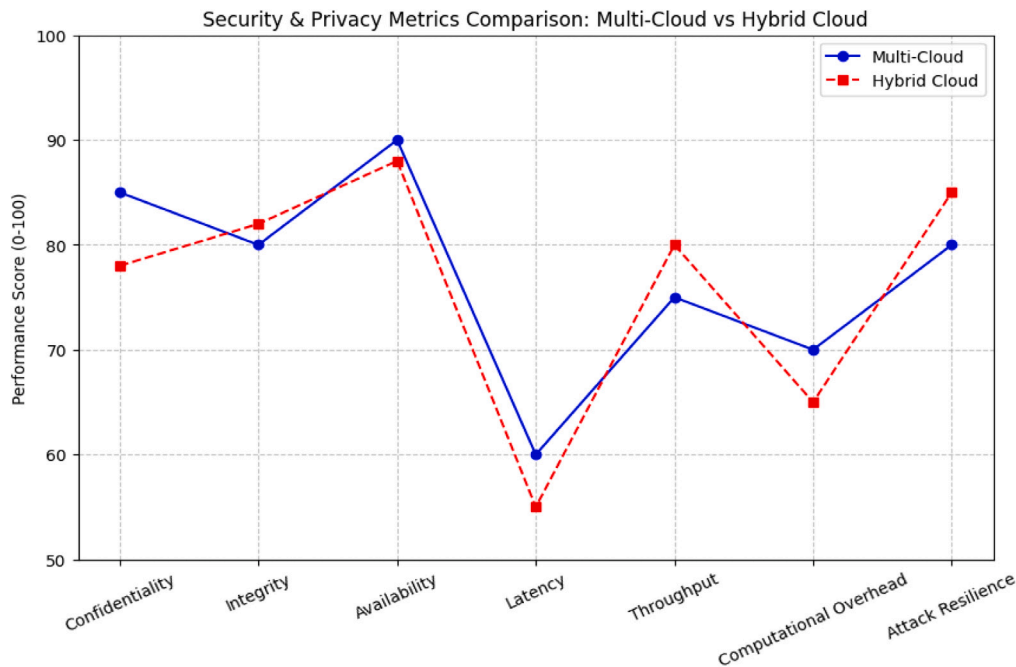


Fig. 3. Security and privacy metrics comparison: Multi-cloud vs. hybrid cloud.

Beyond AI-based solutions, blockchain-based decentralized security architectures are gaining traction as promising approaches to enhance trust, transparency, and reliability in cloud security. With inherent characteristics such as distributed ledgers, consensus mechanisms, and cryptographic validation, blockchain technology can address many of the risks associated with centralized security systems. By distributing security controls, blockchain minimizes single points of failure and mitigates the consequences of compromised nodes, an advantage particularly relevant in multi-cloud and hybrid cloud scenarios. Furthermore, blockchain can enable secure identity management, enforce access control, and provide transparent audit trails, ensuring that sensitive cloud resources are only accessible to authorized parties.

Another prominent development is the integration of Privacy-Preserving AI for Secure Cloud Services, which aims to reconcile the demand for data-driven AI insights with the imperative of protecting user privacy. With AI models more dependent on the data, the problem of data exposure and its utilization in the wrong way has emerged. Federated learning, differential privacy, and homomorphic encryption help organizations analyze distributed datasets without directly processing raw data. These technologies are very useful, especially in areas of human sensitivity, such as health, finance, and government, which require higher levels of data security.

Finally, the interaction of 5G and Edge Computing Security and Multi-Cloud Infrastructures, which by itself depicts a potential revolution of cloud security platforms. The arrival of 5G networks and Edge computing that aims at transferring data processing and data storage closer to the data source, opens up new possibilities in terms of delay and scalability of failures. While beneficial by decentralization, it also increases the risk area of operations, as well as complicating protection in terms of the various types of systems. In such circumstances, it is possible to mitigate risks by developing an integrated security solution that works in association with other established systems. When implemented together with multi-cloud platforms and the edge, both 5G security and edge security need to be integrated and elastic to protect both data and applications effectively.

Collectively, these trends underscore the urgent need for next-generation security strategies that are dynamic, scalable, and resilient in the face of rapid technological change and increasingly sophisticated cyber threats. As cloud adoption continues to reshape organizational workflows, it is anticipated that new structural paradigms

will emerge, including post-quantum cryptography, AI-based adaptive defense mechanisms, decentralized blockchain architectures, privacy-preserving AI models, and edge cloud-integrated security. The developed innovations aim to eliminate current threats and establish sustainable cloud ecosystems, enabling organizations to harness the full potential of cloud computing while maintaining security and privacy.

Cloud environment changes constantly create new security problems which force organizations to establish strong defensive measures to protect their systems from impending risks. Post-quantum cryptography plays a crucial role in safeguarding data against quantum-enabled attacks, while AI-driven self-adaptive security mechanisms facilitate real-time threat detection and mitigation. Furthermore, decentralized security architectures that leverage blockchain enhance data integrity and eliminate a single point of failure. Privacy-preserving AI techniques allow for secure cloud services without compromising user data. Meanwhile, the integration of 5G and edge computing security with multi-cloud environments addresses critical concerns related to latency and scalability. A detailed comparison of these emerging trends is presented in (Table 5).

8. Open challenges and research opportunities

8.1. Scalability of security mechanisms in large-scale multi-cloud environments

The increasing adoption of multi-cloud architectures introduces significant scalability challenges, particularly in implementing robust security mechanisms that can efficiently scale with the expanding cloud ecosystem. Traditional security models, primarily designed for single-cloud deployments, often fall short in addressing the complexities introduced by multi-cloud infrastructures. Since cloud workloads are dynamically shifting from one service provider to another, it becomes very difficult to apply and enforce the same security policies in multiple and different environments. One of the research gaps that needs to be filled is the ability to design decentralized, scalable, and robotic security, that is suitable for cloud space and multiple cloud providers with little performance degradation. For that reason, other distributed IAMs must also grow adapted to federative authentication and authorization models that work well on various cloud infrastructures. Since

Table 5
Comparison of emerging trends and future directions in cloud security.

Emerging trend	Key features	Security benefits	Challenges	Potential applications
Post-Quantum Cryptography (PQC) for cloud security	Quantum-resistant encryption (e.g., lattice-based, hash-based, code-based cryptography)	Protects against quantum computing attacks, ensures long-term data security	High computational complexity, transition challenges from classical cryptography	Cloud data security, financial transactions, government communications
AI-driven self-adaptive security mechanisms	AI/ML-based anomaly detection, automated threat mitigation, adaptive security policies	Real-time threat detection, proactive defense against cyberattacks	Model explainability issues, vulnerability to adversarial AI attacks	Autonomous cybersecurity systems, SOC automation, intrusion detection
Decentralized security architectures using blockchain	Distributed ledger technology, smart contracts, secure identity management	Eliminates single points of failure, enhances data integrity and auditability	Scalability issues, regulatory and compliance challenges	Secure access control, identity verification, decentralized cloud storage
Privacy-preserving AI for secure cloud services	Federated learning, differential privacy, homomorphic encryption	Enables AI-driven insights without exposing raw data, enhances user privacy	High computational overhead, potential trade-offs in model accuracy	Healthcare data analysis, financial fraud detection, AI-driven cybersecurity
Integration of 5G and edge computing security with multi-cloud	Secure communication protocols, distributed security frameworks, low-latency encryption	Enhances security in distributed cloud environments, reduces latency and attack surfaces	Increased complexity, interoperability challenges between multi-cloud and edge networks	Smart cities, industrial IoT, remote healthcare, autonomous vehicles

they are considered to have less effectiveness in security management, future research should focus on improving the decentralization of these frameworks using distributed trust models, such as blockchain. Another great solution is artificial intelligence protection which helps maintain security settings as variables and can learn, manage load balancing, and act based on threat intelligence. However, more research is needed to determine the use of software-defined security (SDS) to implement policies for each cloud, as well as for each roll of work along with compliance with various regulations.

8.2. Optimizing security-cost trade-offs in hybrid cloud deployments

There is a huge debate about hybrid cloud management between aspects of security and cost. They find themselves always in a dilemma of where to balance between the implementation of more effective security measures, which in turn increase operation costs, or the implementation of weak measures since the valued assets are at risk. Selective protection systems have become an area of interest in the recent past, as they offer security protection based on risk assessment and workload values. These should include cost-sensitive models that determine the level of security needed to meet regulatory requirements for each specific workload while at the same time avoiding cost. However, with the help of secure computing technologies such as trusted execution environments (TEEs), for example, the necessary computations can be executed safely and without any considerable burden to the performance. Other potential lines of work include expanding with efficient security-friendly resource provisioning utilizing cost optimization algorithms based on past threats and cloud usage. Therefore, future studies should shed light on the new security-as-a-service (SECaaS) business model, with its concept that allows organizations to outsource their security needs and employ services to pay by usage, depending on the need or workload of the organization.

8.3. Ensuring compliance with evolving data protection regulations

As technology continues to evolve rapidly and the number of rules and regulations related to cross-border data protection increases, compliance in multi-cloud and hybrid cloud environments has become a highly complex issue. Regulations such as the general data protection regulation (GDPR), the California Consumer Privacy Act (CCPA), and various industry-specific mandates impose stringent requirements on data residency, user consent, and data processing practices. A key research gap lies in the development of automated compliance verification frameworks that can continuously evaluate cloud environments for regulatory adherence. These frameworks should incorporate compliance-as-code methodologies, where regulatory policies are embedded directly into cloud infrastructure deployment scripts, ensuring automated and consistent enforcement of compliance measures.

In addition, privacy-preserving cryptographic techniques—such as homomorphic encryption and secure multiparty computation—warrant further exploration to enable compliance-aligned data processing mechanisms that reduce legal risks. Another significant research opportunity involves the creation of cross-jurisdictional data governance frameworks capable of facilitating compliance with conflicting regulatory requirements across different regions. These frameworks should support dynamic policy adaptation in response to regulatory updates and include secure mechanisms for auditing and compliance reporting. Future advancements in AI-driven compliance monitoring also have the potential to transform the way organizations detect and remediate regulatory violations in real-time, thus minimizing legal liabilities while maintaining operational flexibility.

The increasing complexity of multi-cloud and hybrid cloud environments presents several critical security challenges that demand innovative solutions. One of the primary challenges is ensuring the scalability of security mechanisms across large-scale multi-cloud infrastructures, which necessitates the development of decentralized and adaptive security frameworks. In addition, optimizing the trade-off between security and cost in hybrid deployments remains a significant concern, as organizations must balance strong protection with financial limitations. In addition, compliance with rapidly evolving data protection regulations requires the implementation of automated compliance verification frameworks and the adoption of cryptographic techniques that preserve privacy. (Table 6) provides a comprehensive comparison of these open challenges, highlighting key research gaps and potential research directions to effectively address them.

8.4. Research questions and proposed solutions

To guide the reader through the core motivations of this study, (Table 7) is intended to summarize the key research questions addressed in the field along with the corresponding solutions proposed in existing, peer-reviewed literature.

The table outlines key research challenges in multi-cloud and hybrid cloud security, along with high-level solutions proposed in this work. To ensure scalability with performance, it suggests lightweight cryptography, AI-driven adaptive security, and decentralized identity. For balancing security and cost, dynamic access controls and cost-aware encryption are proposed. Compliance with evolving regulations is addressed via AI monitoring, privacy auditing, and blockchain tracking. Secure data sharing is enabled through ABE, SMPC, and homomorphic encryption. Inter-cloud authentication is strengthened using federated identity and zero-trust models. Cross-cloud threats are mitigated by AI-based anomaly detection and TEEs. Real-time threat detection is improved via federated learning and cloud-native analytics. Privacy in machine learning is ensured through differential privacy and secure

Table 6

Comparison of open challenges and research opportunities in cloud security.

Challenge	Research gaps	Potential research directions
Scalability of security mechanisms in large-scale multi-cloud environments	Lack of decentralized and automated security frameworks for multi-cloud setups.	Blockchain-based distributed trust models for scalable security.
	Inconsistent security policies across different cloud providers.	AI-driven adaptive security mechanisms for real-time threat intelligence.
	Inefficient identity and access management across federated environments.	Federated authentication and software-defined security for dynamic policy orchestration.
Optimizing security-cost trade-offs in hybrid cloud deployments	High operational costs of advanced security mechanisms in hybrid clouds.	Cost-aware security models with dynamic resource allocation based on risk.
	Limited adoption of confidential computing techniques due to computational overhead.	Confidential computing (e.g., TEEs) for cost-efficient secure processing.
	Lack of predictive analytics for cost-effective security provisioning.	ML-based predictive analytics for security investment optimization.
Ensuring compliance with evolving data protection regulations	Difficulty in automating compliance verification in dynamic cloud environments.	Compliance-as-code approaches for automated regulatory enforcement.
	Complexities in cross-jurisdictional data governance.	AI-driven regulatory monitoring and automated compliance adaptation.
	Limited privacy-preserving cryptographic techniques for compliance-friendly processing.	Homomorphic encryption and secure multi-party computation for compliance.

Table 7

Research questions and proposed solutions for security and privacy in multi-Cloud and hybrid cloud environments.

Research question	Proposed solution
How can security mechanisms be scaled efficiently in large-scale multi-cloud environments while maintaining performance?	Develop lightweight cryptographic protocols, employ AI-driven adaptive security frameworks, and use decentralized identity management to minimize performance bottlenecks.
What are the optimal trade-offs between security and cost in hybrid cloud environments?	Design dynamic risk-based access control models and cost-aware encryption techniques that balance security, performance, and financial feasibility.
How can compliance with evolving global data protection regulations (e.g., GDPR, CCPA) be ensured in multi-cloud and hybrid cloud deployments?	Implement AI-driven compliance monitoring systems, privacy-preserving auditing mechanisms, and blockchain-based regulatory tracking frameworks.
What strategies can enhance secure data sharing across heterogeneous cloud providers without compromising confidentiality?	Utilize attribute-based encryption (ABE), secure multi-party computation (SMPC), and homomorphic encryption to enable controlled data access across different cloud platforms.
How can secure inter-cloud authentication and access management be standardized to prevent unauthorized access?	Develop blockchain-based federated identity management systems and zero-trust access control models to ensure secure authentication across cloud environments.
What mechanisms can mitigate the risk of cross-cloud attacks, including side-channel attacks and data exfiltration?	Employ AI-driven anomaly detection, confidential computing techniques, and hardware-assisted trusted execution environments (TEEs) to enhance security.
How can real-time threat detection be improved in distributed cloud infrastructures without introducing excessive overhead?	Leverage federated learning-based intrusion detection systems (IDS), AI-driven threat intelligence sharing, and cloud-native security analytics.
What privacy-enhancing techniques can be employed to enable secure machine learning in cloud environments?	Use differential privacy, secure federated learning, and privacy-preserving model training techniques to protect sensitive data during AI-based computations.
How can decentralized approaches enhance trust and security in hybrid and multi-cloud ecosystems?	Implement blockchain-based smart contracts for access control, decentralized identity management solutions, and distributed consensus mechanisms to enhance security.
What strategies can be adopted to optimize encryption efficiency while ensuring strong data security in resource-constrained cloud environments?	Develop hybrid encryption models combining symmetric and asymmetric encryption, employ lightweight cryptographic techniques, and use hardware-assisted encryption acceleration.

model training. Finally, decentralization is supported with smart contracts, and encryption efficiency is enhanced through hybrid models and hardware acceleration.

9. Conclusion

The increasing adoption of multi-cloud and hybrid cloud environments has introduced a new paradigm in cloud computing, offering organizations enhanced flexibility, scalability, and operational resilience. However, this transition has also presented significant security and privacy challenges that must be addressed to ensure the integrity, confidentiality, and availability of critical data. This study highlights key security risks related to data protection, identity and access management, network security, and regulatory compliance in multi-cloud and hybrid infrastructures. The analysis reveals that data security concerns, such as ensuring confidentiality, integrity, and secure distributed storage, remain major obstacles, particularly due to the complexities introduced by data fragmentation across multiple cloud service providers.

Furthermore, the study emphasizes the difficulties in securing inter-cloud authentication and authorization, where inconsistencies in access control models expose systems to potential cyber threats.

A variety of threats exist in multi-cloud environments. Common attack trends identified in the research include man-in-the-middle (MitM) attacks, side-channel attacks, ransomware threats, and data exfiltration. Addressing these threats requires the implementation of security pre-processors, such as encryption techniques, zero-trust security models, and blockchain-based security mechanisms. Moreover, concerns regarding compliance with data protection regulations—such as the general data protection regulation (GDPR), the california consumer privacy act (CCPA), and other jurisdiction-specific legal frameworks—continue to pose significant challenges for enterprises adopting multi-cloud infrastructures. With the advancement of technology and the enforcement of global data privacy laws, organizations are now required to develop comprehensive compliance frameworks capable of monitoring regulatory landscapes in real-time while minimizing the risks associated with cross-border data handling.

Finally, this study defines the open issues and future research directions mentioned extensively in the literature. Recent years have shown that cloud-based security measures may not be as effective for large-scale systems, and thus a scalable solution to address it is still needed. Potentially, the introduction of AI-driven threat detection and the concept of a blockchain-based trust model could be seen as promising research directions. Similarly, to balance the risks involved and the costs associated with it, enhanced paradigms of hybrid cloud computing with adequate security measures that are vigorous but feasible are needed. Therefore, there is a need to advance verification and compliance testing techniques, as well as cryptographic approaches to protect personal data according to existing and emerging laws.

9.1. Recommendations for cloud security research and development

In the case of security and privacy in multi-cloud and hybrid clouds, the future course of action should be aimed at the development of a new security architecture that integrates new technologies with practical and feasible implementation models. That is why the first recommendation is to design AI and ML-based security orchestration solutions to assess threats, exclude anomalies, and apply security controls. These should support automated, risk-based security management and reduce the use of manual configurations as much as possible, thus also enhancing the reaction time to new threats.

An important research direction is enhancing cross-cloud IAM services through decentralized identity management and blockchain-based trust models to strengthen user-to-host authentication. Further studies should explore advanced cryptographic techniques like homomorphic encryption, secure multiparty computation (SMPC), and confidential computing to enable secure data processing without exposing sensitive information. Additionally, regulatory concerns should drive the development of policy-based cloud architectures by integrating compliance-as-code into deployment strategies. The future of AI-driven compliance tools will also be vital, allowing organizations to self-monitor and ensure data protection and regulatory adherence.

The integration of blockchain and DLT with multi-cloud security architecture also offers the potential to improve accountability, traceability, and reliability among cloud participants. Some of the research agendas should focus on creating new reliable security models that utilize blockchain to enhance distributed identity authentication, secure and immutable operational event logs, and provable compliance records.

Therefore, security is a moving target due to the new and complex multi-cloud and hybrid cloud environments of today, and in the face of these changes and increasing regulations, one has to keep up with the pace of change constantly. By integrating AI-driven security automation, advanced cryptographic techniques, and policy-aware cloud management frameworks, future research and development can significantly improve the resilience and trustworthiness of cloud computing ecosystems. Collaboration among organizations, cloud service providers, and the research community will be crucial to driving advancements in cloud security, ultimately fostering a safer, more efficient, and regulation-compliant digital infrastructure.

CRedit authorship contribution statement

Sijjad Ali: Visualization, Validation, Supervision, Software, Resources, Project administration. **Dhani Bux Talpur:** Writing – review & editing, Writing – original draft, Visualization, Data curation, Conceptualization. **Adeel Abro:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Khulud Salem Alshudukhi:** Writing – review & editing, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Ghadah Naif Alwakid:** Funding acquisition, Formal analysis, Conceptualization, Methodology. **Mamoona Humayun:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Farhan Bashir:**

Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shuaib Ahmed Wadho:** Methodology, Formal analysis, Data curation, Conceptualization. **Asadullah Shah:** Visualization, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The corresponding author hereby declares that there are no conflicts of interest in connection with this research work. He has no financial, personal, or other relationships with individuals or organizations that could inappropriately influence or be perceived to influence the content of this manuscript. Furthermore, He affirms that all sources of funding for this study have been acknowledged, and the study was conducted with funding body. Author involved in this research have disclosed any potential conflicts of interest, and none were identified. The review research presented in this manuscript is an original work and has not been previously published nor is it currently under consideration for publication elsewhere.

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Data availability

No data was used for the research described in the article.

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